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JGR Atmospheres

RESEARCH ARTICLE
10.1029/2024JD042080

Key Points:

- Summertime carbonaceous aerosol concentrations in the Interior North Slope of Alaska (NSA) were twice as high as those on the coast
- Carbonaceous aerosol was predominately organic carbon, with at least 86% of the total carbon (TC) coming from contemporary sources
- Wildfire in North American boreal forests was linked to TC concentrations that were 3–13 times higher than during background periods

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Welch, A. M., Matthews, T., Sheesley, R. J., Wang, H., Barsanti, K. C., Nielsen, N., et al. (2025). Summertime carbonaceous aerosols in interior versus coastal Northern Alaska. *Journal of Geophysical Research: Atmospheres*, *130*, e2024JD042080. https://doi.org/10.1029/2024JD042080

Received 29 JUL 2024
Accepted 31 MAR 2025

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Summertime Carbonaceous Aerosol in Interior Versus Coastal Northern Alaska

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Abstract Rapid warming is likely increasing primary production and wildfire occurrence in the Arctic. Projected changes in carbonaceous aerosols during the summer will impact atmospheric chemistry and climate, but our understanding of these processes is limited by sparse observations. Here, we characterize carbonaceous aerosol in Alaska, USA: Toolik Field Station in the Interior and the Atmospheric Radiation Measurement facility at Utqiaġvik on the Arctic coast, during the summers of 2022 and 2023. We estimated $PM_{2.5}$ and PM_{10} concentrations using laser light scattering (PurpleAir sensors) and examined total carbon (TC) and its organic carbon (OC) and elemental carbon (EC) fractions in total suspended particles (TSP). We investigated the dominant sources of carbonaceous aerosol using air mass backward-trajectories from the NOAA HYSPLIT model and radiocarbon source apportionment of TC. TC concentrations were about twice as high in the Interior compared to the coast, with contemporary sources dominating at both Toolik (95%–99%) and Utqiaġvik (86%–89%) over minor contributions from fossil sources. Elevated PM, TC, OC, and EC concentrations coincided with major boreal forest fire activity in North America that brought smoke to the region. The radiocarbon signature of EC measured at Toolik during these wildfire events indicated that over 90% of the EC came from contemporary sources. Our measurements demonstrate the potential for Arctic aerosol concentrations to respond significantly to climate warming-induced changes to the landscape and emphasize the need for continuous atmospheric monitoring to advance our understanding of this rapidly changing environment.

Plain Language Summary The Arctic is warming rapidly, likely leading to increased plant growth and more wildfires. These changes are expected to affect the particles in the air during the summer. However, our understanding of these processes remains incomplete due to limited observations, despite some previous studies at these sites and at the pan-Arctic level. In this study, we measured the amount and composition of particles in the air at two locations in Alaska: Toolik Field Station and Utqiaġvik, during the summers of 2022 and 2023. We found that the amount of carbon-containing particles was higher in the Interior of Northern Alaska than on the coast. By combining computer models of wind patterns with laboratory analyses of the composition of the particles, we also found that most of the particles came from natural sources, specifically compounds that are emitted by plants as well as wildfires, while only a small amount came from the combustion of fossil fuels. Our findings show that climate change, through rising temperatures and more favorable conditions for wildfires, may influence the quantity and composition of particles in the summer Arctic air, underscoring the need for monitoring to understand these changes and their impact on air quality and climate.

1. Introduction

With mineral dust and sea spray, carbonaceous aerosol plays an important role in modulating the Arctic's energy budget and precipitation patterns (Post et al., 2019). In the clean Arctic atmosphere, carbonaceous aerosol serves as cloud condensation nuclei (CCN), affecting the formation and properties of clouds and, in turn, impacting surface radiation and precipitation (X. Li et al., 2020; Schmale et al., 2021). Total carbonaceous aerosol is commonly separated into organic carbon (OC) (which can be further separated into non-refractory OC and refractory OC, or brown carbon (BrC)) and light-absorbing black or elemental carbon (EC). OC accounts for 10%–40% of the total particulate mass and most of the carbonaceous mass, with similar concentrations during winter and summer (approximately 0.1–1.4 μg m^{-3} throughout the pan-Arctic) (Moschos, Dzepina, et al., 2022). EC concentrations are 1–2 orders of magnitude lower (annual average 28 ± 24 ng C m^{-3}) and show a strong seasonal trend, with higher concentrations in the winter between October and June (Winiger et al., 2019). Upon deposition,

light-absorbing EC changes the albedo of snow and ice-covered surfaces, which further alters the surface energy budget. EC concentrations across the Arctic have declined by about 20% since the 1980s in response to decreasing emissions from local Arctic and non-Arctic European and Russian sources (Ren et al., 2020). In contrast, summertime OC emissions are expected to increase since climate change is rapidly transforming the landscape (Post et al., 2019) via longer fire and growing seasons and increased area of open water, which lead to higher emissions of particles and aerosol precursors into the atmosphere (Abbatt et al., 2019; Schmale et al., 2021). Our ability to assess and forecast the Arctic's changing atmosphere, however, is limited by a lack of observations during the summer.

Much of our knowledge of the Arctic's atmosphere is centered on the phenomenon of Arctic Haze which is controlled by emissions originating from lower latitudes (Fisher et al., 2010; Law & Stohl, 2007; Mori et al., 2020; Quinn et al., 2007). The Arctic atmosphere can be described as a dome of cold air that is encompassed by the polar front (Barrie, 1986) which expands into lower latitudes during the winter. Subsequent poleward transport of anthropogenic pollutants emitted from urban and industrial centers in the mid-latitudes, along with the limited scavenging that happens during the polar night, leads to the buildup of air pollutants in the High Arctic in winter and spring. During the summer, the dome shrinks and limits the transport of pollutants from mid-latitudes, while emissions from local sources are more efficiently scavenged during the polar day, leading to a more pristine boundary layer (Hamilton et al., 2014). Clean air policies have decreased the transport of pollutants to the Arctic during the winter (Quinn et al., 2008; Sharma et al., 2004; Winiger et al., 2019). However, there is evidence of increased emissions in South and East Asia leading to increases in some aerosol concentrations (Ren et al., 2020), as well as the potential for increased aerosol production due to the increase in the marginal ice zone in the summer (Schmale et al., 2021).

Carbonaceous aerosols in the Arctic originate from both anthropogenic and natural sources, which are undergoing significant transformations alongside trends in fossil fuel emissions and climate change. Major anthropogenic sources of carbonaceous aerosols in the Arctic include fossil fuel combustion and industrial emissions, including those from Arctic oil fields (Gunsch et al., 2017) and those transported from lower latitudes (Willis et al., 2018). While pollutant transport has declined since 1989 (Mackie et al., 2016); increased shipping activity in the region could raise local emissions and ozone concentrations (Dalsøren et al., 2013; Peters et al., 2011; Schmale et al., 2018; Zhan et al., 2014). Future emissions and particulate matter (PM) from fossil sources remain largely uncertain and dependent on future policy decisions (Schmale et al., 2018). Wildfires and secondary organic aerosols (SOA) are the major natural sources of carbonaceous aerosol in the Arctic. Wildfire in boreal forest and tundra biomes has increased in severity and frequency with climate change (Chen et al., 2021; Hu et al., 2015; Kasischke & Turetsky, 2006; McCarty et al., 2021). Wildfire smoke has been shown to elevate OC and EC concentrations by 1–2 orders of magnitude above ambient levels, producing highly oxidized monomers from the degradation of cellulose, hemicellulose, and lignin along with some polycyclic aromatic hydrocarbons and their derivatives (e.g., resinous acids) (Mouteva, Fahrni, et al., 2015; Schneider et al., 2024). SOA formation is changing due to increased biogenic volatile organic compound (BVOC) emissions, driven by longer growing seasons and warmer temperatures (Kellomäki et al., 2001; Kramshøj et al., 2016; Lindwall, Schollert, et al., 2016; Lindwall, Svendsen, et al., 2016; Rinnan et al., 2020; Valolahti et al., 2015; Vettikkat et al., 2023; H. Wang, Welch, Nagalingam, Leong, Czimczik, et al., 2024). Arctic BVOC emissions—dominated by isoprene and, to a lesser extent, monoterpenes (Sindelarova et al., 2014; Tang et al., 2023)—are expected to increase further following the northward advance of the treeline (Tang et al., 2023) as well as the observed expansion of broadleaf deciduous boreal trees (Rinnan et al., 2020) and deciduous Arctic shrubs (Simin et al., 2022; H. Wang, Welch, Nagalingam, Leong, Kittitananuvong, et al., 2024).

Though significant changes in summertime aerosol concentrations in the near-surface Arctic atmosphere are anticipated, models continue to have substantial difficulty representing atmospheric chemistry of the Arctic. This difficulty is primarily due to a lack of fundamental information and understanding around trace gas and aerosol emissions sources, aerosol composition and chemistry, and their interactions with climate (Christian et al., 2017), all stemming from the fact that models are primarily based on the results of studies conducted in temperate regions. For example, the Interagency Monitoring of Protected Visual Environments (IMPROVE) network, which operates 158 sites across the U.S., has only two sites in Arctic Alaska (>60°N), of which only one is in the tundra biome. The limited data available indicate that there are significant differences in atmospheric transport processes, chemistry, and emissions, demonstrating the need for more direct observations in the Arctic.

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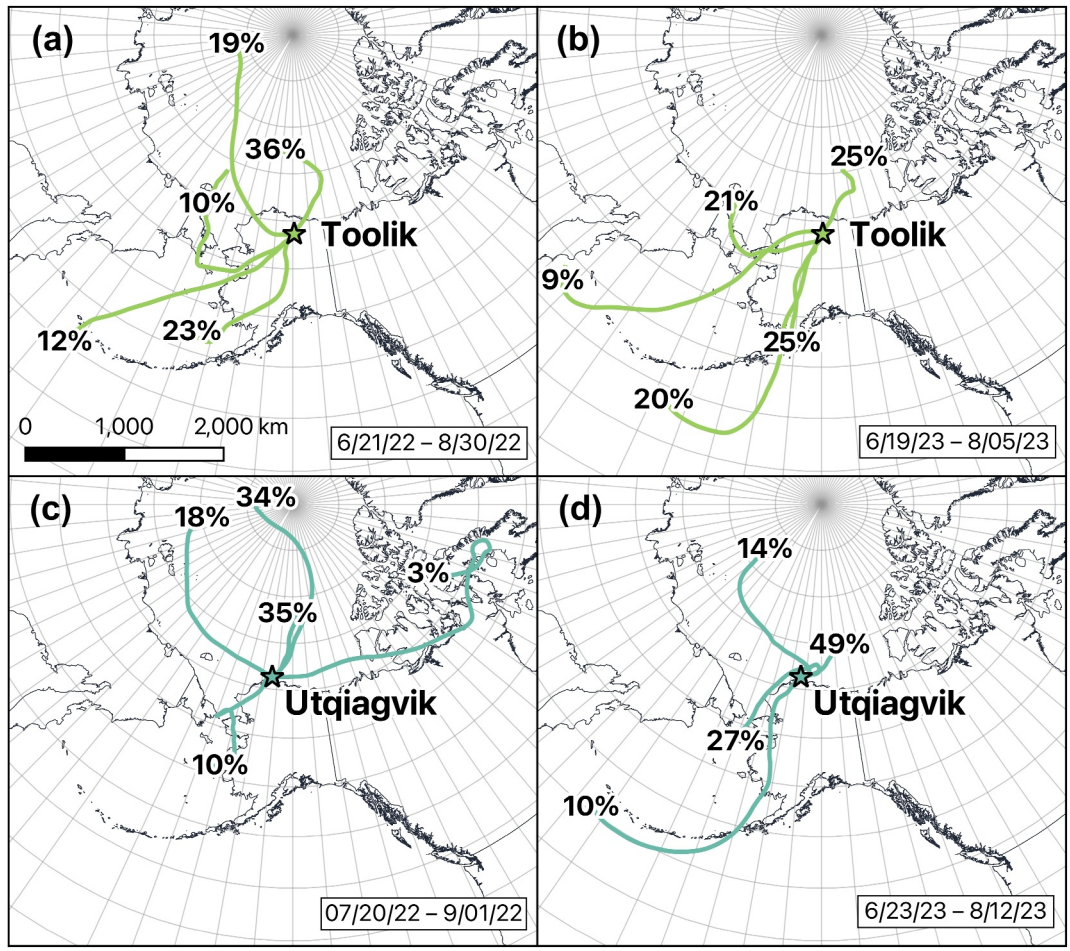

Figure 1. Air mass contributions at Toolik, (a), (b) and Utqiaġvik, AK, USA (c), (d) during the summer of 2022 and 2023, respectively (NOAA HYSPLIT v. 5.2, 10-day, 6 hr, starting height 10 m agl, default top of model (10 km agl)).

While contemporary organic aerosol constitutes a major proportion of the Arctic's summertime aerosol burden, its sources and impacts on atmospheric chemistry and climate remain poorly understood (Moffett et al., 2022; Moschos, Dzepina, et al., 2022; Winiger et al., 2019). In Alaska, previous observations have largely focused on the boreal region (Deshmukh et al., 2019; Mouteva, Czimczik, et al., 2015) and the northern coast (Barrett & Sheesley, 2017; Quinn et al., 2009), while observations from the Interior are sparse (Creamean et al., 2018). Therefore, we characterize carbonaceous aerosol concentrations in northern Alaska during the summer of 2022 and 2023, based on weekly collections of total suspended particles (TSP) and continuous monitoring of particulate matter concentrations with PurpleAir PA-II sensors at Utqiaġvik on the Arctic coast and Toolik Field Station in the interior of the North Slope of Alaska (NSA). We quantify the sources and composition of carbonaceous aerosol by combining modeled air mass backward trajectories, thermal-optical analysis, and radiocarbon (^{14}C) source apportionment of TC.

2. Materials and Methods

2.1. Site Description

TSP samples were collected at two sites on the NSA in the North American Arctic: Toolik Field Station (68.63°N, 149.60°W) and Utqiaġvik (71.29°N, 156.79°W) (Figure 1) during the summer (JJA) of 2022 (n = 17) and 2023 (n = 16). Toolik Field Station—established, operated, and managed by the University of Alaska, Fairbanks in cooperation with Division of Polar Programs, Directorate for Geosciences at the National Science Foundation (NSF)—is located north of the Brooks Range (600 km north of Fairbanks, AK), where *moist dwarf-shrub, tussock-graminoid* (A.K.A. *moist acidic*) tundra is the dominant ecosystem type (Muller et al., 1999). Utqiaġvik is

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located at the northernmost point of Alaska on the coast of the Arctic Ocean, where *moist graminoid, prostrate-shrub* (A.K.A. *non-acidic*) and *wet graminoid tundra* dominate. Samples at Toolik were taken at the "Flux Hut" location (68.62574°N, 149.60334°W), about 250 m SW of the station camp. This location was chosen to minimize the potential impact from local, contemporary sources of particulate matter originating from field station activity (e.g., biomass burning at the sauna and waste incinerator). Most wind at this location during the study period was N, S, or NW and under 10 m/s (Figure S1 in Supporting Information S1). Samplers at Utqiaġvik were deployed at the Department of Energy Atmospheric Radiation Measurement (DOE ARM) facility (71.323°N, 156.615°W), which has been determined to have minimal contribution from the Utqiaġvik village (Barrett et al., 2015). Samples were taken on top of the Radar Wind Profiler Instrument Platform, about 10 m asl.

2.2. Data Collection

TSP were collected using a high-volume sampler (Hi-Q Environmental Products Company, Inc., San Diego, CA, USA) onto quartz fiber filters (520 cm^2 in total size, Pall Corporation, Port Washington, NY, and GE Healthcare Technologies, Inc., Chicago, IL, USA) with an average flow rate of 1.1 m^3 min^{-1}. Sampling duration was approximately 7 days (6.4 ± 2.2 days), with deviations for capturing pollution events (i.e., wildfire smoke) or due to logistical challenges. Prior to sampling, filters were pre-combusted at 450°C for 4 hr and stored in wrapped aluminum foil inside a zip-lock bag. For ^{14}C analysis, blank filters were attached to the high-volume concurrent with samples for the entire sample duration but not exposed to flow. For thermal-optical OC/EC analysis, field blanks were mounted onto the sampler (not exposed to flow) and then removed. After sample collection, filters were kept frozen and transported to the University of California, Irvine (UCI) and Baylor University for laboratory analyses.

Particulate Matter (PM_{10} and $PM_{2.5}$) mass concentrations (μg m^{-3}) were measured by laser particle counting using PurpleAir PA-II air quality sensors (PurpleAir, Draper, UT, USA map.purpleair.com) and averaged over 24-hr intervals. Data from two sensors that were away from point sources at Toolik (one at the same location as the high-vol TSP sampler and one atop the dining hall of the field station) were used. One sensor was used at Utqiaġvik, which was situated in the town. Uncalibrated measurements from the two channels on each sensor (a and b) were averaged. Note that these data were not calibrated against other standardized methods; however, measurements from these sensors have been found to be in reasonable agreement with reference measurements (Stavroulas et al., 2020).

The historical record of OC and EC concentrations at Toolik (available since 2018) was analyzed by the Interagency Monitoring of Protected Visual Environments (IMPROVE) network and downloaded using the IMPROVE Data Q/A and Validation Site-Level Analyses tool (https://views.cira.colostate.edu/fed/Express/ImproveQA.aspx) (Malm et al., 1994).

2.3. Laboratory Analyses

2.3.1. TC Analysis

TSP filters were analyzed for total carbon (TC) mass (μg C cm^{-2}) independently using thermal-optical transmittance (TOT) as the sum of the OC and EC fractions (see Section 2.3.2) and during ^{14}C analysis from each sample's CO_2 yield (see Section 2.3.3). Both methods were used to obtain a final estimate of TC because both methods analyze only portions of the whole filter, on which loading may not be completely homogeneous. All TC measurements were blank-corrected, normalized by filter area used during each analysis, and averaged. The blank for both Toolik and Utqiaġvik samples was 0.2 ± 0.1 μg C cm^{-2} (mean ± SD, $n = 18$) and 0.2 ± 0.2 μg cm^{-2} ($n = 23$) using TOT or vacuum line extraction during ^{14}C analysis, respectively. Flow rate and sampling duration were then used to calculate TC in μg C m^{-3} air. TC was blank-corrected and an uncertainty of combusted CO_2 yield of 5% was assumed to account for filter measurement and instrument uncertainty. The method detection limit for TC measured during graphitization and combustion was 1 μg (Mouteva, Fahrni, et al., 2015).

2.3.2. OC/EC Analysis

OC and EC mass (μg C cm^{-2}) were measured with the NIOSH 5040 method (Birch & Cary, 1996) using a TOT carbon analyzer (Sunset Laboratories, Tigard, OR) along with instrument blanks and sucrose standards at Baylor University. OC and EC of field blanks were measured using the same method and averaged per sampling site and

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year, then subtracted from sample values. Flow rate and sample volume were then used to calculate OC and EC fractions in ng C m^{-3}. Analytical triplicates were done for each batch of 10 samples, which were also run alongside instrument blanks and a sucrose standard. The method detection limit (MDL) was 5.5 ng m^{-3} for OC and 0.37 ng m^{-3} for EC.

2.3.3. Isotopic Analysis

For filters with aerosol loading above the detection limit for elemental analysis (50 μg TC per punch)—89% of Toolik samples; 0% Utqiaġvik samples—two 1.5 cm^2 punches were folded into tin capsules (5 mm × 9 mm) and analyzed for %C and $\delta^{13}C$ using an elemental analyzer (Fisons NA-1500NC, Thermo Scientific, Waltham, MA, USA) coupled with an isotope ratio mass spectrometer (DeltaPlus XL, Thermo). The measurement uncertainty for $\delta^{13}C$ based on long term secondary standard measurements was 0.1‰.

Radiocarbon analysis was performed at the W. M. Keck Carbon Cycle Accelerator Mass Spectrometry (KCCAMS) laboratory at UC Irvine. For ^{14}C measurements of TC, a portion of each filter (equivalent to 100–300 μg C) was evacuated and sealed in a pre-combusted quartz tube (6 mm (≤12 1.5 cm^2 punches) or 9 mm (≤36 1.5 cm^2 punches)) along with 80 mg of CuO catalyst and combusted at 900°C for 3 hr. One sample (UCIAMS# 286654) was combusted in Pyrex at 550°C for 4 hr due to its high salt content, which causes quartz tubes to crack during combustion; thus, the measured $\Delta^{14}C$ of that sample is only representative of the OC fraction since EC will not combust at the lower temperature. We also quantified the ^{14}C signature of the EC fraction of the four samples with the highest TC load (all from Toolik), three from 2022 and one from 2023. Following the extraction of water-soluble OC, the EC mass from three 1.5 cm^2-filter punches was isolated using the Swiss_4S protocol on a modified Sunset OC/EC analyzer and combined on a vacuum line (Mouteva, Czimczik, et al., 2015; Zhang et al., 2012).

Then, the produced CO_2 was isolated cryogenically on a vacuum line. TC-CO_2 was sealed in a pre-combusted Pyrex reaction tube along with reagents of Zn and TiH_2, and catalyst (Fe), and reduced to graphite at 550°C for 4 hr following (Xu et al., 2007). Blanks and EC samples were graphitized using a modified sealed-tube zinc reduction method for ultra-small samples (Walker & Xu, 2019). $\Delta^{14}C$ of the graphite was measured using accelerator mass spectrometry alongside graphite produced from reference materials (Beverly et al., 2010). The measurement uncertainty of $\Delta^{14}C$ was <3‰ from processing standards and blanks (Xu et al., 2007). $\Delta^{14}C$ relates the measured $^{14}C/^{12}C$ of the sample to the $^{14}C/^{12}C$ of the National Institute of Standards and Technology Oxalic Acid I (OX-I) standard according to Equation 1:

$$\Delta^{14}C = \frac{\left.\frac{^{14}C}{^{12}C}\right]_{\text{sample},-25}}{\left.\frac{^{14}C}{^{12}C}\right]_{\text{OX-I},-19}\, e^{\frac{(y-1950)}{8267}}} \tag{1}$$

where $^{14}C/^{12}C$] sample, −25 refers to the isotope ratio normalized to a $\delta^{13}C$ of −25‰, $^{14}C/^{12}C$] OX-I, −19 refers to the isotope ratio of OX-I normalized to a $\delta^{13}C$ of −19‰, y is the year of sampling, and 8,267 is the mean ^{14}C lifetime (Stuiver & Polach, 1977). $\Delta^{14}C$ is the commonly reported nomenclature for modern geochemical studies, as it corrects for isotopic fractionation of ^{14}C as well as the decay of OX-I standard since 1950 (conventionally, "present"). To correct for the extraneous carbon on the filters, $\Delta^{14}C$ measurements were blank-corrected using a mass-balance approach (Equations 2 and 3):

$$m_{obs} = m_b + m_{smp} \tag{2}$$

$$\Delta^{14}C_{smp} = \frac{\Delta^{14}C_{obs}m_{obs} - \Delta^{14}C_b m_b}{m_{obs} - m_b} \tag{3}$$

where $\Delta^{14}C_{smp}$, $\Delta^{14}C_{obs}$, and $\Delta^{14}C_b$, represent the true $\Delta^{14}C$ of the sample, the observed $\Delta^{14}C$ of the sample and blank mixture, and the blank, respectively. m_{obs} and m_b represent the observed and blank mass concentrations (TC), respectively. Error in blank-corrected $\Delta^{14}C$ values was obtained by propagating field blank $\Delta^{14}C$ error and measured $\Delta^{14}C$ error (Hwang & Druffel, 2005; Odwuor et al., 2023) (see Text S1 in Supporting Information S1).

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All samples were analyzed alongside size-matched standard materials and corrected for contamination by extraneous carbon during laboratory processing (Walker & Xu, 2019).

2.3.4. Radiocarbon Source Apportionment

Contemporary (modern) and fossil sources of TC and EC in bulk TSP were apportioned using another isotope mass balance (Equations 4 and 5):

$$m_{smp} = m_{ff} + m_m \tag{4}$$

$$\Delta^{14}C_{smp} = (1 - m_m)\,\Delta^{14}C_{ff} + m_m\Delta^{14}C_m \tag{5}$$

where m_{ff} is the fossil and m_m is the contemporary contribution to the total carbon mass of the sample. $\Delta^{14}C_{ff}$ is −1,000‰, the known signature of fossil fuels. The contemporary end member ($\Delta^{14}C_m$) varies according to the type of contemporary sources. During clean air conditions, we chose the mean ^{14}C of atmospheric CO_2 measured in Utqiaġvik in the summer of 2022, which was −6‰. When sites were impacted by wildfire smoke (identified using the National Oceanic and Atmospheric Administration (NOAA)'s Hazard Mapping System (Section 2.5)), we estimated $\Delta^{14}C_m$ using a Keeling Plot approach (Figure S2 in Supporting Information S1) (Keeling, 1958; Odwuor et al., 2023; Pataki et al., 2003) (Equation 6):

$$\Delta^{14}C_{fire+bkgd} = m_{bkgd}\left(\Delta^{14}C_{bkgd} - \Delta^{14}C_{fire}\right)\left(\frac{1}{m_{fire+bkgd}}\right) + \Delta^{14}C_{fire} \tag{6}$$

in which the measured Δ^{14}C of each sample is a mixture of wildfire smoke and well-mixed background ($\Delta^{14}C_{bkgd+fire}$), m_{bkgd} and $m_{fire+bkgd}$ are the mass concentrations (TC) of the background and measured background and wildfire smoke mixture, respectively, and $\Delta^{14}C_{bkgd}$ and $\Delta^{14}C_{fire}$ are the Δ^{14}C of the background and smoke, respectively. A type II ordinary least squares regression—which accounts for random error in x ($1/m_{fire+bkgd}$) and y ($\Delta^{14}C_{fire+bkgd}$), as neither was controlled—was fit to a selection of measurements from Toolik using the "lmodel2" package in R (Legendre, 2018; R Core Team, 2023) (Figure S2 in Supporting Information S1). Measurements were selected based on dominant air mass source (south/boreal zone) determined from air mass backward trajectories (Section 2.4). Thus, $\Delta^{14}C_{fire}$, which was used as the mean boreal forest fire smoke Δ^{14}C end member for apportionment, was the intercept of the linear model, 58‰ (95% CI [23, 93]) and substituted for $\Delta^{14}C_m$ in samples affected by wildfire smoke events in Equation 5.

2.4. Air Mass Backward Trajectories

Ten-day air mass backward trajectories were calculated at Toolik and Utqiaġvik using the NOAA HYSPLIT model, version 5.2 (January 2022) (Rolph et al., 2017; Stein et al., 2015). Meteorological data from the Global Data Assimilation System (GDAS) provided by NOAA was used as input for the model. For each TSP sample, trajectories were calculated every 6 hr for the sample's duration at Toolik and Utqiaġvik with a starting height of 10 m above ground level and a default top of model (10,000 m agl) (Figure S3–S6 in Supporting Information S1). Ten-day trajectories were also calculated every 6 hr for the sampling campaign's duration in 2022 and 2023 shown in Figure 1. The backward trajectories were then clustered in HYSPLIT by combining trajectories until there was a marked increase in the total spatial variance. The clusters were exported and mapped using QGIS 3.18 (QGIS Development Team, 2023). Air mass backward trajectories were used to identify major source regions for TSP samples. Previous work demonstrated that HYSPLIT outputs are suitable for such use, though the limited number of meteorological stations in the Arctic limits accuracy (Barrett et al., 2015).

2.5. Data Access and Statistical Methods

Temperature records during JJA 2022 and 2023 for correlation with TC and OC measurements were downloaded from the University of Utah MesoWest (https://mesowest.utah.edu) (Toolik = TUAA2 Station; Utqiaġvik = PABR Station). The following shapefiles were used to create Figure 1: World International Boundaries (Center, 2013) and 5-Degree Graticule Grid, World, 1:110 million, 2012 (Kelso & Patterson, 2012).

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A Student's *t*-test was performed in R to compare concentrations and $\Delta^{14}C$ among sites, years, and wildfire smoke events/background periods. TC, OC, and EC concentrations were log-transformed and $PM_{2.5}$ and PM_{10} were square root-transformed prior to *t*-tests to meet the assumption of residual normality. TC and PM_{10}, OC, EC, and ambient temperature were tested for correlation using the Mann-Kendall method in R. Ambient temperature and OC concentrations were correlated using an ordinary least squares regression. OC and EC concentrations were correlated using a type II ordinary least squares regression using the lmodel2 package in R (Legendre, 2018; R Core Team, 2023). This method accounts for random error in both the x and y variables; additionally, it is recommended that the significance of the slope estimate be tested by permutation ("p-perm") (Legendre, 2018).

The Hazard Mapping System (HMS) Smoke Data Statistics from NOAA was used to quantify days during the campaign when smoke was observed in the total atmospheric column above Toolik or Utqiaġvik (NOAA, 2017). For each day, smoke presence was treated as a binary (i.e., smoke or no smoke) and the relative number of smoke days during each TSP sample's duration was calculated as "Percent Smoke Days." "Light Smoke" periods were designated as sample durations which experienced ≤25% smoke days at the sampling site. This threshold was chosen by considering at what percentage of smoke days TC, OC, and EC, increased significantly more than measurement uncertainty (Figure S7 in Supporting Information S1).

Data were analyzed and figures were made using R version 4.3.0 and the tidyverse package (Wickham et al., 2019) in R studio (R Core Team, 2023).

3. Results and Discussion

Our TSP sampling design in Alaska during the summers of 2022 and 2023 captured a wide range of aerosol source regions and atmospheric conditions. Toolik received primarily terrestrial and Utqiaġvik marine air masses (Figure 1). Based on analyses of HYSPLIT backward trajectories, the majority of air masses reaching Toolik (60%–75%) arrived from regions to the S/SW, W or N/NW and were transported across the boreal and/or the tundra regions of Alaska and Russia's Far East, with only about a third of air masses originating from the N/NW (Beaufort Sea). Conversely, most air transported to Utqiaġvik (60%–90%) originated from the N/NW and traveled across the Chukchi or Beaufort Sea, with lesser contributions from the S/SW (10%–30%, Bering Sea) and E (<5%, Canadian Archipelago). Furthermore, most of our study period represented typical summer conditions, but we also captured pollution episodes at both locations. 44% of sampling days at Toolik and 33% at Utqiaġvik were influenced by heavy wildfire smoke based on NOAA HMS. Since all prior ground observational studies of carbonaceous aerosol on the NSA were done on the Arctic coast (most commonly, in Utqiaġvik), our sampling approach allowed us to capture surface TSP representative of Arctic Alaska and allowed unprecedented insight to TSP originating from terrestrial sources.

3.1. Aerosol Concentrations and Composition

Continuous measurement of PurpleAir PM at both sites showed that summertime aerosol concentrations were comparable inland at Toolik and on the coast at Utqiaġvik, except during wildfire smoke events, when concentrations were elevated at Toolik (Figure 2). Background conditions were similar, supported by comparable median PM_{10} at the two sites (0.3 ± 10 μg m^{-3} at Toolik compared to 0.5 ± 2.3 μg m^{-3} at Utqiaġvik). However, mean PM_{10} and $PM_{2.5}$ were consistently greater at Toolik compared to Utqiaġvik (t = 3.9(degrees of freedom = 226); p = 1e-04 and t = 4.3(228); p = 2e-05, respectively), indicating significant influence from pollution. TC, OC, and EC were also all significantly higher at Toolik than Utqiaġvik (t = 3.3(24.6); p = 0.003, t = 3.5(24.6); p = 0.002, and t = 3.2(13.5); p = 0.007, respectively) (Table 1).

Background PM levels at both sites indicate Good Air Quality Index (AQI) by the U.S. Environmental Protection Agency (EPA) standards (EPA, 2003), with moderate air quality during wildfire periods at Toolik. Considering the entire observation period at each site, the maximum PM_{10} at Toolik was 142 μg m^{-3} (Moderate AQI) on 7/1/2022. At Utqiaġvik maximum PM_{10} was only 16 μg m^{-3} (Good AQI) on 7/5/2022. Mean $PM_{2.5}$ was 85% of mean PM_{10} at Toolik and 72% of PM_{10} at Utqiaġvik, suggesting most of PM was $PM_{2.5}$ on average.

Trends in TC were like that of PM_{10} (Figures 2a and 2b), also indicating significant influence from high concentration events at Toolik. Median TC at Toolik was 0.8 ± 2.1 μg m^{-3} (Figure 2a) and median TC at Utqiaġvik was 0.2 ± 0.3 μg m^{-3} (Figure 2b). TC did not change significantly from 2022 to 2023 at either site. When both sites were sampled concurrently for TSP (i.e., excluding 6/21/22–7/19/22 at Toolik (when there was no sampling

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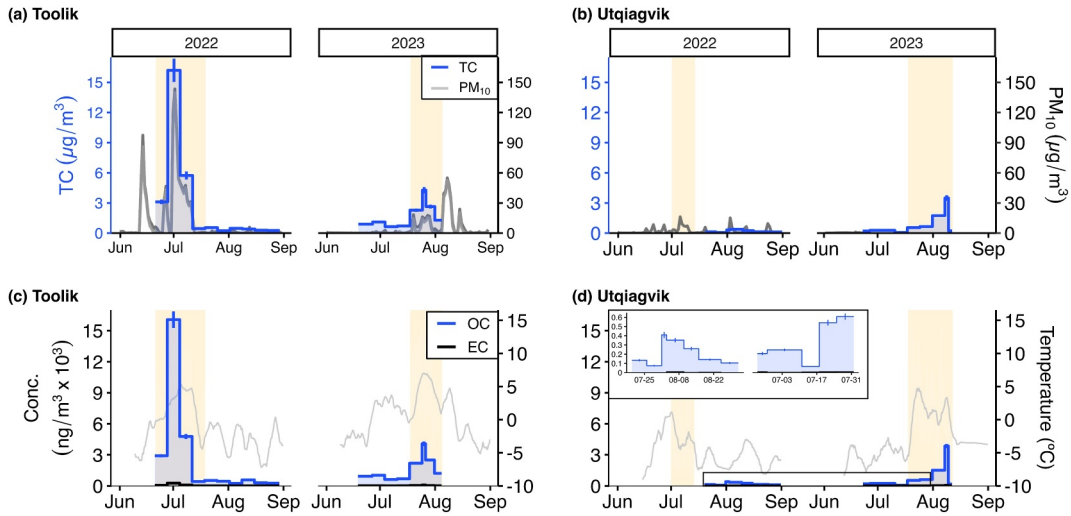

Figure 2. Aerosol concentrations and composition and air temperature on the North Slope of Alaska (NSA), USA during the summer of 2022 and 2023, with periods of wildfire smoke as observed by NOAA HMS shaded in yellow. Total carbon (TC) (thermal-optical transmittance or combustion) and PM_{10} (PA-II, Purple Air) (a), (b) and mean 5-day air temperature and concentration of organic carbon (OC) and elemental carbon (EC) (thermal-optical transmittance) at Toolik and Utqiaġvik, respectively (c), (d). The inset in panel (d) shows OC and EC concentrations measured from 7/15/2022 to 9/1/2022 and 6/15/2023 to 8/1/2023 with a smaller concentration range compared to the main plot.

Table 1
The Mean (± Standard Deviation) Concentrations of Total Carbon (TC) (Averaged From Vacuum Line and Thermal-Optical Transmittance Measurements), Organic Carbon (OC), Elemental Carbon (EC), OC/EC Ratios, and $\Delta^{14}C$ of TC During the Summers of 2022 and 2023 at Toolik and Utqiaġvik, AK, USA

	Toolik		Utqiaġvik	
	2022	2023	2022	2023
	$n = 10$	$n = 8$	$n = 7$	$n = 8$
Mean TC (μg m^{-3})	**3 ± 5**	**2 ± 1**	**0.2 ± 0.1**	**0.9 ± 1**
Max TC	16 ± 1	4 ± 0.3	0.4 ± 0.03	4 ± 0.3
Min TC	0.2 ± 0.03	0.6 ± 0.05	0.06 ± 0.01	0.1 ± 0.008
Mean TC–Light HMS Smoke	0.4 ± 0.1	0.8 ± 0.2	0.2 ± 0.1	0.2 ± 0.1
Mean TC–HMS Smoke Events	5 ± 6	2 ± 1	NA	1 ± 1
Mean OC (ng m^{-3})	**2,700 ± 4,900**	**1,700 ± 1,200**	**210 ± 130**	**900 ± 1,300**
Max OC	16,000 ± 150	4,100 ± 220	400 ± 33	3,900 ± 200
Min OC	240 ± 17	620 ± 35	75 ± 9	67 ± 5
Mean EC (ng m^{-3})	**49 ± 84**	**32 ± 30**	**2 ± 4**	**19 ± 17**
Max EC	270 ± 17	97 ± 18	11 ± 4	55 ± 15
Min EC	5 ± 4	8 ± 4	0.01 ± 4	3 ± 2
Mean OC/EC	**45 ± 13**	**58 ± 15**	**7,000 ± 8,800**	**46 ± 30**
Max OC/EC	62	82	24,000	87
Min OC/EC	21	42	33	6
Mean $\Delta^{14}C$ (‰)	**12 ± 40**	**−24 ± 47**	**−147 ± 85**	**−77 ± 88**
Max $\Delta^{14}C$	77 ± 2	29 ± 2	−49 ± 7	65 ± 2
Min $\Delta^{14}C$	−38 ± 4	−106 ± 2	−281 ± 5	−194 ± 2

Note. TC, OC, EC, OC/EC, and $\Delta^{14}C$ maximum and minimum measurements are reported with propagated measurement uncertainties. Mean measurements are bolded.

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at Utqiaġvik) and 8/8/23–8/12/23 at Utqiaġvik (vice versa)), TC, OC, and EC were also all significantly higher at Toolik than Utqiaġvik (t = 3.3(24.6); p = 0.003, t = 3.5(24.6); p = 0.002, and t = 3.2(13.5); p = 0.007, respectively).

Absolute magnitudes and trends in OC closely resembled trends in TC, indicating that almost all of TC is OC (Figure 2). OC concentrations ranged from 96% to 99% of TC at Toolik and from 86% to 100% at Utqiaġvik (when both OC and TC were measured using TOT (Section 2.3.2)). Median OC at Toolik was 810 ± 1,900 ng m^{-3} (Figure 2c) and median OC at Utqiaġvik was 250 ± 340 ng m^{-3} (Figure 2d). High OC/TC ratios (>0.9) are typical for Arctic aerosol year-round (Moschos, Dzepina, et al., 2022; Mouteva, Czimczik, et al., 2015).

During background periods (light HMS smoke), temperature predicted OC concentrations. OC concentrations were linearly correlated with ambient temperature during background periods at Toolik and Utqiaġvik (R^2 = 0.45, p = 0.03 and R^2 = 0.58, p = 0.02, respectively) (Figures 2c and 2d). About 50% and 60% of variation in OC concentrations during background periods can be explained by temperature at Toolik and Utqiaġvik, respectively, a relationship which has been previously reported for Utqiaġvik (Moffett et al., 2022). This relationship points to local/regional BVOC emissions as potential precursors of background OC. The OC fraction may also be affected by long-range transport and accumulation during dry periods (Szidat et al., 2004).

EC made up as much as 14% and as little as <0.01% of TC. Median EC at Toolik for both years was 15 ± 36 ng m^{-3} (Figure 2c) and median EC at Utqiaġvik was 6.8 ± 10 ng m^{-3} (Figure 2d). Low EC/TC ratios are typical in the Arctic during the summer (<0.1) but can be higher near industrial sites (Caiazzo et al., 2021; Moffett et al., 2022; Rodríguez et al., 2020). EC at Toolik did not change significantly from 2022 to 2023, but at Utqiaġvik, EC was significantly higher in 2023 compared to 2022 (t(7.0) = −3.8; p = 0.007) (Table 1).

While temperature drove background aerosol concentrations, wildfire smoke events were a major predictor of anomalously high concentrations, especially at Toolik. Peaks in PM_{10}, TC, OC, and EC concentrations coincident with HMS wildfire smoke events (shown in yellow shading in Figures 2a–2d) were evident at both sites, though timing of TSP sampling at Utqiaġvik did not include late June–early July, when there was a significant smoke event captured at Toolik. For Utqiaġvik, the duration of smoke observed by HMS was longer than the period of high OC observed at the surface, which may indicate that the smoke aloft does not always mix down to the surface at the coastal site. TC, OC, and EC concentrations were all significantly higher during wildfire smoke events (t(19.3) = −5.0; p = 8e-05, t(18.7) = −4.6; p = 2e-04, and t(26.0) = −4.3; p = 2e-04, respectively). At its maximum, TC from wildfire smoke was 16 ± 1.2 μg m^{-3}, 7-times the mean at Toolik. Measured TC reached a maximum at Utqiaġvik of 3.5 ± 0.3 μg m^{-3} in mid-August 2023, 6-times the Utqiaġvik mean and once again coincident with wildfire smoke as observed by HMS. TC at Toolik was not sampled in mid-August 2023, but PM_{10} reached a 2023 maximum during this time of 24 μg m^{-3}. These wildfire smoke events coincided with significant wildfire activity in Alaska in 2022 (Alaska Fire Service (AFS) Alaska Wildland Fire Information Map Series, n.d.) and Canada in 2023, where unprecedentedly severe wildfires emitted an estimated 647 Tg C (570–727 Tg C) (5–30x the norm of the previous decade) (Byrne et al., 2024).

TC was significantly, positively correlated with PM_{10}, OC, EC, and ambient temperature (Figure 3). The correlation between TC and PM_{10} (tau = 0.5, $p << 0.05$) indicates that the carbonaceous fraction is an important component of total aerosol <10 μm (Figure 3a). The correlation between TC and OC (tau = 0.9, $p << 0.05$) (Figure 3b) and TC and EC (tau = 0.8, $p << 0.05$) (Figure 3c) indicates that there are consistent sources of both OC and EC, particularly among the samples affected by wildfire smoke events. Finally, the correlation between TC and ambient temperature (tau = 0.7, $p << 0.05$) points to the potential for BVOCs—the emission of which is temperature-dependent—to be a source of carbonaceous aerosol (similar to the correlation between OC and ambient temperature) (Figure 3d).

Compared to a previous study of summer OC and EC concentrations in TSP at Utqiaġvik between 2015 and 2017 (Moffett et al., 2022), mean OC concentrations in this study were almost 3 times and EC concentrations almost 2 times greater. Mean OC at Toolik was greater than at any site on the NSA sampled before, at 2,200 ± 3,700 ng m^{-3}, but mean EC at Toolik was comparable to that measured at Oliktok Point, Alaska in the aforementioned study period (42 ± 64 ng m^{-3} compared to 43 ± 1 ng m^{-3}). The wide range of OC concentrations observed in this study supports significant contribution from wildfire smoke events.

The implications for climate (i.e., the radiative feedback) associated with carbonaceous aerosol in the Arctic are driven by concentration as well as the chemical properties of the aerosol. While contemporary PM dominates in

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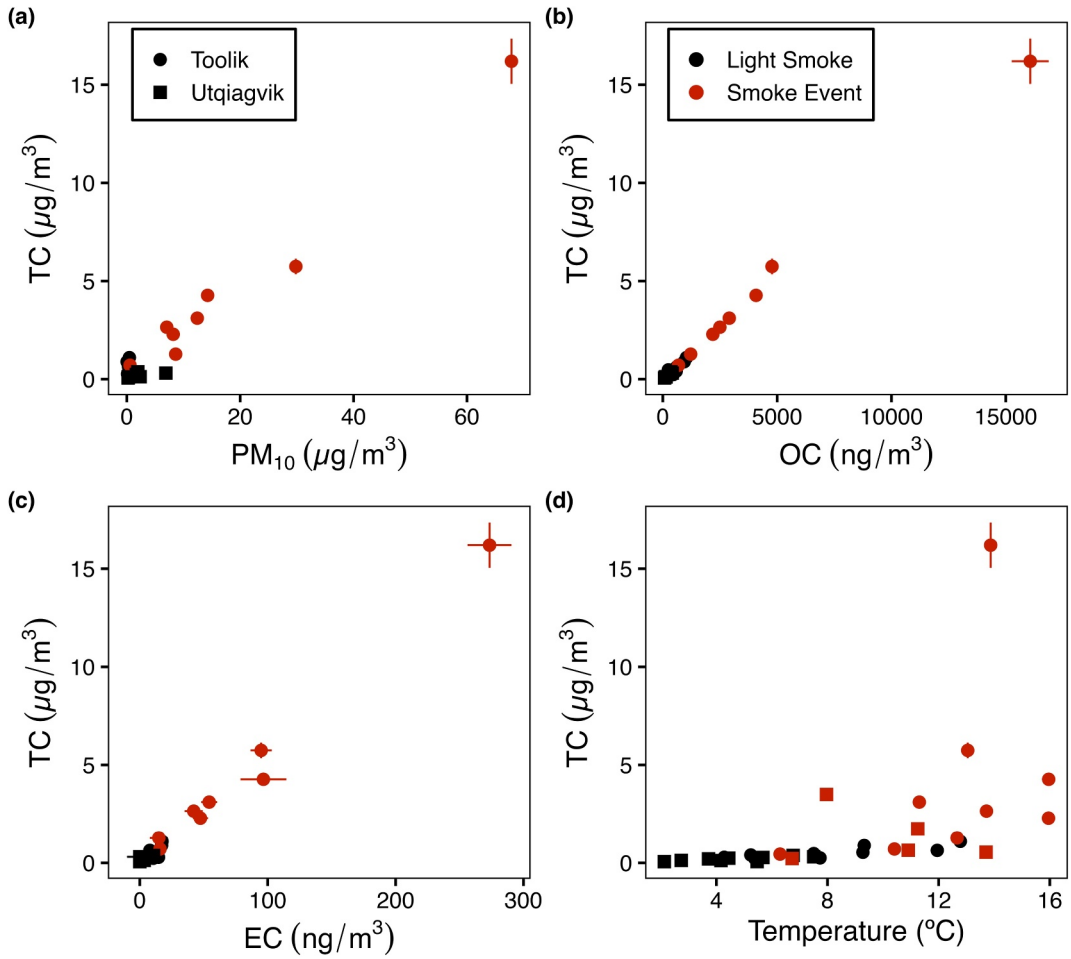

Figure 3. Total carbon (TC) versus PM_{10} (a), organic carbon (OC) (b), elemental carbon (EC) (c) and mean temperature of TSP sample duration (d) at Toolik and Utqiaġvik on the North Slope of Alaska (NSA), USA during the summer of 2022 and 2023. Background samples and samples that were affected by HMS wildfire smoke events are designated by color. Error bars represent measurement uncertainty of TC, OC, and EC.

the summer, overall PM concentrations are higher in the winter and spring (Arctic Haze), in part due to the northward migration of the Arctic front (Quinn et al., 2007; Stone et al., 2014; Winiger et al., 2019). Year-round studies of Arctic PM consistently find elevated concentrations during the winter, predominately from fossil and biofuel combustion in Europe and Siberia (Barrett & Sheesley, 2017; Eleftheriadis et al., 2009; Fisher et al., 2010; Koch & Hansen, 2005; Moschos, Schmale, et al., 2022; Sharma et al., 2004, 2006). Wintertime EC concentrations are high across the Arctic, with high contribution from fossil sources (Barrett et al., 2015; Barrett & Sheesley, 2017; Manousakas et al., 2020; Rodríguez et al., 2020). Wintertime PM—both OC and EC fractions—tends to be depleted in ^{14}C compared to summertime (Barrett & Sheesley, 2017; Rodríguez et al., 2020). Summertime aerosol tends to be more light scattering, in part due to the higher OC fraction relative to winter/spring, when contribution from EC increases light absorption (Quinn et al., 2002). It is estimated that, globally, aerosols reduce the magnitude of warming by greenhouse gas emission by 1/3 (J. Li et al., 2022); however, direct radiative forcing from carbonaceous aerosols alone has been estimated to be 0.65 W m^{-2} (Chung et al., 2012). It is also estimated that changes in vegetation composition (namely, the northward expansion of boreal needleleaf species) will increase aerosol optical depth (i.e., increase the redirection of sunlight by aerosols) (Tang et al., 2023). Furthermore, organic particles may serve as more effective cloud condensation nuclei, as fresh soot is hydrophobic (Sun & Ariya, 2006). Thus, high concentrations of OC could alter cloud formation processes. The high OC contribution to TC found in this study would indicate that surface carbonaceous aerosol in summertime Northern Alaska is largely light scattering, thereby cooling the surface, particularly during background periods. However, without the characterization of the light-absorbing BrC fraction of OC, the net effect of carbonaceous aerosols on

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the radiative budget is unclear, especially during wildfire smoke events. A recent study found that boreal forest fire aerosol emissions were a positive feedback for Arctic warming, potentially outweighing decreases in EC (Zhong et al., 2024). Our analysis suggests that inland areas may be especially vulnerable to experiencing high concentrations during smoke events.

3.2. Aerosol Sources

The ratio of primary or secondary OC to primary EC and the isotopic signatures of TC provide additional insight into the origin of summertime TSP in the near-surface atmosphere on the NSA. Radiocarbon analyses of carbonaceous aerosol are particularly powerful when performed on the EC fraction because EC is a primary aerosol that originates only from combustion processes. However, most of the samples analyzed in this study were too small (<65 μg C cm^{-2}) for ^{14}C analysis of the EC fraction alone and thus we focused on ^{14}C analysis of TC. Radiocarbon analyses of EC are, however, presented for selected samples with the highest loading, which were taken during wildfire smoke events at Toolik.

3.2.1. OC/EC Ratios

We found high OC/EC compared to other NSA studies. OC/EC range from 21 to 82 at Toolik and from 6–>>1,000 at Utqiaġvik. There was a vast range in OC/EC values between sites and sampling years (Table 1). In 2022, mean OC/EC was 45 ± 13 at Toolik and 7,000 ± 8,800 at Utqiaġvik. In 2023, mean OC/EC was 58 ± 15 at Toolik and 46 ± 30 at Utqiaġvik. During 2022, some Utqiaġvik samples had very low EC concentrations, resulting in highly skewed OC/EC values. High OC/EC ratios (>10) are typical for Arctic aerosols during the summer, except in gas-flaring regions (Manousakas et al., 2020; Moffett et al., 2022; Rodríguez et al., 2020).

Furthermore, OC and EC concentrations were significantly correlated at both Toolik and Utqiaġvik (R^2 = 0.99; permuted p-value (p-perm) = 0.01 and R^2 = 0.75; p-perm = 0.01, respectively), indicating that both originated from the same source (Figure S8 in Supporting Information S1). Notably, however, the correlations were heavily influenced by pollution from wildfire smoke. During light HMS smoke periods, the correlation between OC and EC was still significant at Toolik (R^2 = 0.50; p-perm = 0.03), but not at Utqiaġvik. It should be noted that Toolik experienced some amount of light smoke in the total column throughout most of the 2023 season, even during background periods, outside of significant wildfire smoke events. The correlation between OC and EC concentrations is also strong when the majority of air is transported from S/SW (R^2 = 0.98; p-perm = 0.01), compared to when it was transported from N/NW (R^2 = 0.73; p-perm = 0.04) (Figure S9 in Supporting Information S1).

Considering samples from both sites and years, the slope of EC as a function of OC—57—is many times greater than the range of 1.1–2.1 that has been reported for urban aerosol (Castro et al., 1999; Lim & Turpin, 2002; Szidat et al., 2004). This slope is similar at Toolik (57) and Utqiaġvik (55). Compared to an urban environment with a higher density of combustion sources, a high relative OC concentration is expected in the Arctic, considering the importance of local sources in the summer and the lack of long-range transport (Moffett et al., 2022).

3.2.2. Isotopes

Isotopic analysis suggests that, while both sites have aerosol depleted in radiocarbon, this depletion is more extreme at Utqiaġvik, suggesting a greater proportion of contemporary sources inland. This signature is reversed during wildfire events, when aerosol is enriched in radiocarbon compared to atmospheric CO_2. For Toolik, TSP was slightly enriched in ^{13}C in 2023 compared to 2022 (no $\delta^{13}C$ data were available for Utqiaġvik) (Figure S10 in Supporting Information S1). $\delta^{13}C$ ranged from −27.7 to −26.2‰ in 2022 and from −27.2 to −24.4‰ in 2023, with no significant difference in means of −26.9 ± 0.4‰ and −26.3 ± 0.9‰, respectively ($t(10.1) = -1.7$, $p = 0.1$). The median value across both sampling years is −26.8 ± 0.7‰, which is typical for vegetation dominated by C3 plants. The $\delta^{13}C$ of Arctic C3 plants ranges from −25 to −35‰, depending on species (Kristensen et al., 2011).

Toolik TC was consistently enriched in ^{14}C compared to that at Utqiaġvik ($t(19.8) = 4.1$; p = 5e-4) (Figures 4a and 4b). $\Delta^{14}C$ values above the current atmospheric CO_2 signature (Figure 4, dashed line) indicate more recent C sources (<60 years old), while lower values suggest older, fossil-derived C. Mean $\Delta^{14}C$ of TC at Toolik was −4 ± 46‰ and mean $\Delta^{14}C$ at Utqiaġvik was −110 ± 91‰. At Toolik, $\Delta^{14}C$ ranged from −38 to 77‰ in 2022

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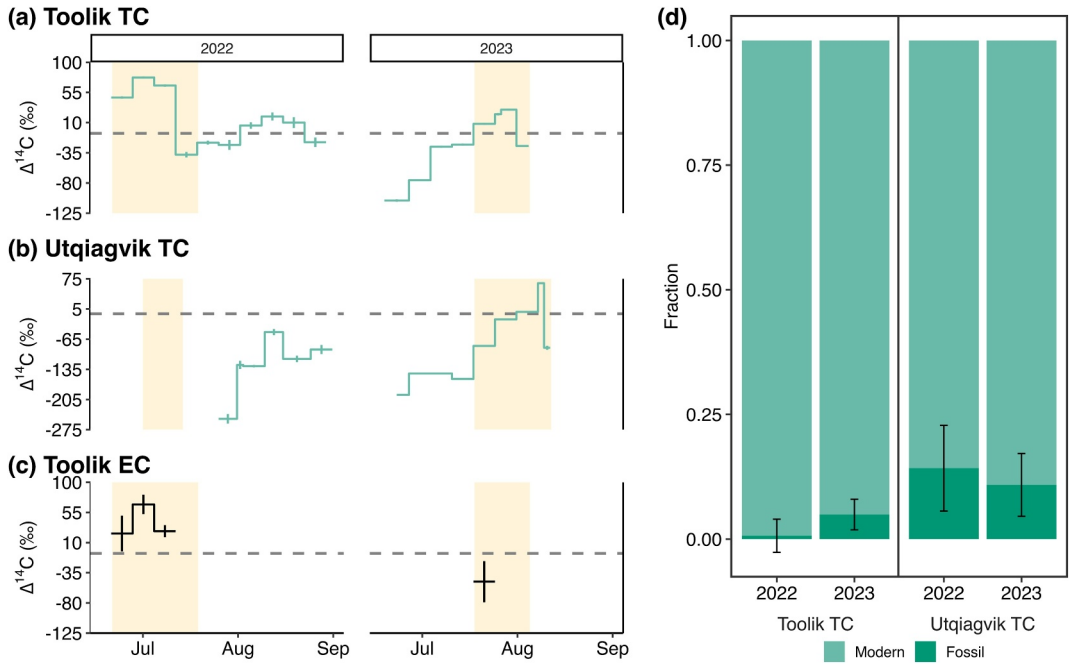

Figure 4. Isotopic composition and source apportionment of carbonaceous aerosol on the North Slope of Alaska (NSA), USA, during the summer of 2022 and 2023, with periods impacted by wildfire smoke shaded in yellow. Radiocarbon signature of total carbon (TC) at Toolik and Utqiaġvik, respectively (a), (b) and of elemental carbon (EC) at Toolik during selected periods with high loading (c). The gray dashed line represents the $\Delta^{14}C$ of atmospheric CO_2 at time of sampling. Vertical error bars represent propagated error induced from blank corrections. Mean relative contributions of contemporary (modern) and fossil sources of TC at Toolik and Utqiaġvik for both summers (d). Error bars represent standard deviations.

and from −106 to 29‰ in 2023. At Utqiaġvik, $\Delta^{14}C$ ranged from −281 to −49‰ in 2022 and from −194 to 65‰ in 2023.

TC at both sites tended to be depleted in ^{14}C compared to ambient CO_2 in 2022 (gray dashed line in Figures 4a–4c) (X. Xu, Pers. Comm. 2024). High $\Delta^{14}C$ values (enriched above ambient CO_2) corresponded with wildfire smoke events (yellow shading in Figures 4a–4c). Using a Keeling plot approach, the mean $\Delta^{14}C$ of TC emitted from biomass burning in North America and transported to the NSA was estimated to be 58‰ (Figure S2 in Supporting Information S1). This estimate is relatively consistent with the mean $\Delta^{14}C$ of $PM_{2.5}$ (103 ± 23‰) from 2013 Alaskan boreal forest fires found in a study by Mouteva, Czimczik, et al. (2015), when the $\Delta^{14}C$ of atmospheric CO_2 at Utqiaġvik (Pt. Barrow) was 23 ± 3‰ (about 20‰ higher than when this study was conducted).

$\Delta^{14}C$ of EC during the 2022 wildfire smoke event at Toolik ranged from 24 ± 27‰ to 67 ± 14‰ ($n = 3$) and was −48 ± 31‰ ($n = 1$) during the 2023 wildfire smoke event at Toolik (Figure 4c). These measurements are elevated by 350‰ on average compared to a mean Arctic summer $\Delta^{14}C$ of EC measured at Alert, Canada in 2014 (−361.8 ± 162.7‰) (Rodríguez et al., 2020). This $\Delta^{14}C$ range constitutes 90%–100% biomass burning contribution to EC (using a biomass burning $\Delta^{14}C$ end member of 58‰ (see Section 2.3.4)). In a previous year-round, circum-Arctic study, Utqiaġvik (Barrow)—representing the NSA—had the lowest biomass burning contribution to EC of all the Arctic monitoring sites, though a summer month from 6/13–7/12 is missing from the Utqiaġvik sampling record that is not missing from the sampling record at other sites (Winiger et al., 2019). Biomass burning was considered an insignificant contributor to aerosol for this 2012–13 study at Utqiaġvik due to orographic lifting at the Brooks Range and the persistent marine boundary layer (Winiger et al., 2019). However, this was not the case at Toolik during the summers of 2022 and 2023 nor at Utqiaġvik in the summer of 2023, as observed in this study. The comparison of TC average concentrations for 2016–17 (Moffett et al., 2022) and 2022 versus 2023 also shows the episodic influence of wildfire smoke at Utqiaġvik, with a much higher maximum TC associated with the wildfire event in August 2023. Toolik had very different influences from wildfire in 2022 and 2023, which highlights the different transport patterns at the inland versus coastal sites on the NSA.

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Using ^{14}C source apportionment to estimate contemporary and fossil contributions to TC, a large majority of TC at both Toolik and Utqiaġvik was found to be contemporary, though the fossil fraction is slightly lower at Toolik compared to Utqiaġvik (Figure 4d and Figure S11 in Supporting Information S1). Contemporary carbonaceous aerosol may include PM from biomass burning (i.e., smoke) or primary or secondary biogenic organic aerosol. Carbonaceous aerosols of fossil origins include soot and other particulates from incomplete combustion of fossil fuels. The large oil field at Prudhoe Bay, AK, USA is a potential source of these aerosols of fossil origins, as previous studies found influence of the Prudhoe Bay oil field on atmospheric particles in Utqiaġvik (Gunsch et al., 2017) and regionally on the NSA (Creamean et al., 2018). The mean fractions for each site and year are as follows: 2022 Toolik 99 ± 3% contemporary and 1 ± 3% fossil, 2023 Toolik 95 ± 3% contemporary and 5 ± 3% fossil, 2022 Utqiaġvik 86 ± 9% contemporary and 14 ± 9% fossil, 2023 Utqiaġvik 89 ± 6% contemporary and 11 ± 6% fossil. The choice in contemporary end member ($\Delta^{14}C$ of contemporary CO_2 or $\Delta^{14}C$ estimated for biomass burning) did not result in a significant change in the contemporary contribution fraction (6%), but it does result in a significant relative change in fossil contribution fraction (86%), since the fossil contribution fraction is so small.

Though the dominance of contemporary OC in the summer is consistent in the literature (Moschos, Dzepina, et al., 2022; Winiger et al., 2019; Yttri et al., 2023), prior studies have found that contribution to be around 63%–74% on the NSA (Barrett & Sheesley, 2017; Moffett et al., 2022). Contemporary carbon comprised 95%–99% and 86%–89% of carbonaceous aerosol at Toolik and Utqiaġvik, respectively (Figure 4d). Thus, TSP samples from Toolik tended to be more ^{14}C-enriched (−6 ± 46‰, on average) than TSP samples from Utqiaġvik (−110 ± 91‰) ($t(19.9) = 4.0, p = 7e{-}4$) (Figures 4a and 4b). The high proportion of contemporary carbonaceous aerosol has several possible attributions. Most likely, the influx of wildfire smoke during the study period has a large role. Increases in the emission of SOA precursors (biogenic organic volatile compounds, BVOCs), local differences (i.e., proximity to industrial activity), and implementation of fossil fuel regulations are also possible drivers.

The high proportion of contemporary contribution could be bolstered by declining contributions from fossil sources as the world transitions away from fossil fuels. After the historically high levels of Arctic Haze in the 1970s–80s, particulate concentrations fell following the collapse of the Soviet Union, coincident with new power plant regulations in the U.S. and Europe (Quinn et al., 2008; Sharma et al., 2004). Though midlatitude emissions are a major contributor to Arctic pollutants, low-altitude, long-range transport is weakened, and wet scavenging is strengthened in the summer; thus, local sources become more important (Garrett et al., 2010; Law & Stohl, 2007; Popovicheva et al., 2019; Quinn et al., 2007, 2008; Stone et al., 2014; Willis et al., 2018). Concentrations of carbonaceous air pollutants—namely, EC—in the Arctic atmosphere decreased in the two decades following 1990 (Sharma et al., 2013), though another study failed to find a consistent trend during the summer, specifically (Schmale et al., 2022). But now, international interest in oil and gas exploration—particularly deep-sea drilling—has sparked following the loss of Arctic sea ice (Brutschin & Schubert, 2016; Intergovernmental Panel on Climate Change (IPCC), 2022). It is estimated that about 30% of the world's oil and gas reserves are in the Arctic (Gautier et al., 2009; Shapovalova, 2020). This interest, as well as the predicted increase in shipping activity (Corbett et al., 2010; Zhan et al., 2014) will increase local sources of EC alongside the expected rise in boreal forest fires (Balshi et al., 2009; de Groot et al., 2013; Flannigan et al., 2005; Gillett et al., 2004), making the understanding and quantification of current sources and concentrations important for future work.

3.2.3. Source Regions

Samples with higher concentrations of OC and EC also tended to have higher $\Delta^{14}C$, indicating relatively young sources of wildfire smoke events (Figure 5). Figure 5 shows that $\Delta^{14}C$ of TSP increases logarithmically as OC increases ($R^2 = 0.6, p << 0.05$) and that EC generally increases with both $\Delta^{14}C$ and OC. There were clear patterns according to site and major source region. Utqiaġvik aerosol tended to be depleted in $\Delta^{14}C$, OC, and EC compared to Toolik. Additionally, the clean air at Utqiaġvik tended to come mostly from the N/NW (Beaufort Sea, Arctic Ocean, Chukchi Sea, Northern Pacific Ocean), whereas more air at Toolik was from the S/SW, that is, traveled across the Alaskan boreal zone. Finally, samples with the highest $\Delta^{14}C$, OC, and EC corresponded to HMS wildfire smoke events and persistent smoke presence on the NSA (red-ringed samples in Figure 5). This relationship points to wildfire as a regional, if episodic, driver of carbonaceous aerosol on the NSA, as wildfire impacts are dependent on consistency in boreal events and transport. This finding is consistent with previous studies which have found boreal forest fires during previous wildfire seasons to impact aerosol concentrations and

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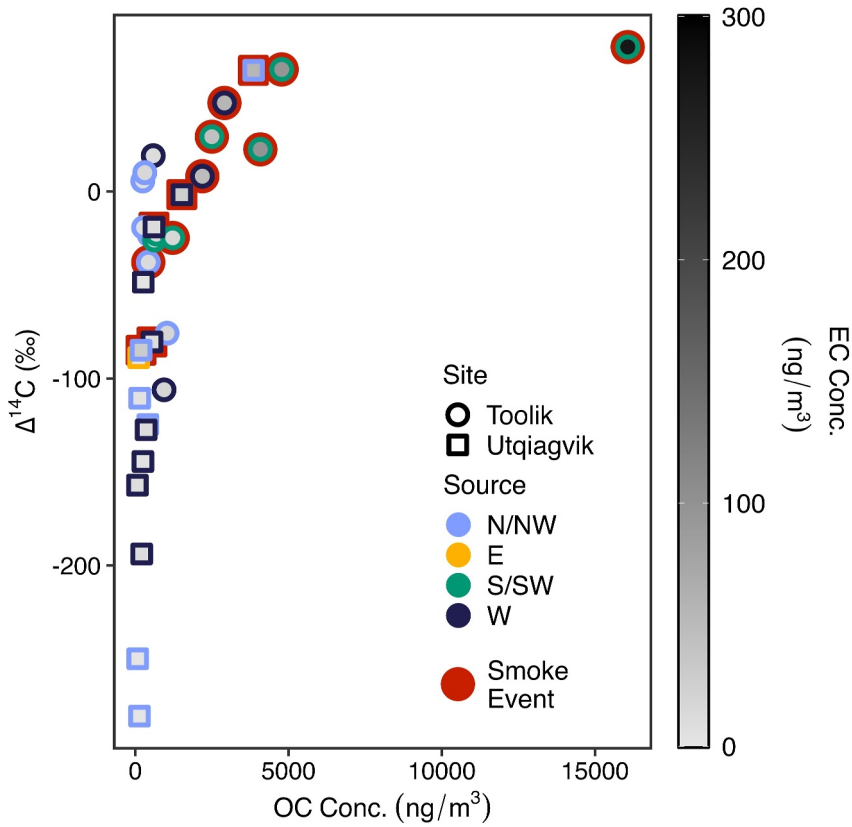

Figure 5. Δ^{14}C versus organic carbon (OC) of total suspended particles (TSP) sampled at Toolik (circles) and Utqiaġvik (squares). Elemental carbon (EC) concentration is represented by shading of each point. The major source of air to each sample is represented by the outline color. Samples that were affected by wildfire smoke events determined using NOAA Hazard Mapping System (HMS) are represented by an additional red outline.

composition on the NSA at the regional scale (Creamean et al., 2018; Moffett et al., 2022). However, due to the elevated concentrations at Toolik, the stronger linear relationship between OC and EC (relative to Utqiaġvik), and the increased air mass source contribution from the south (relative to Utqiaġvik) (Figure 1, Figures S3 and S4 in Supporting Information S1), wildfire smoke may be a stronger determinate of TC concentrations inland than on the Arctic coast. Net losses in aboveground carbon have been linked to intensifying boreal forest fires (Walker et al., 2019; J. A. Wang et al., 2021; Xu et al., 2024), which are predicted to increase in frequency and severity with climate change (Balshi et al., 2009; de Groot et al., 2013; Flannigan et al., 2005; Gillett et al., 2004), and will have cascading ramifications for summer atmospheric chemistry in the historically "pristine" Arctic.

Together, our data shows the high impact that wildfire events can have on summertime aerosol concentrations of the lower troposphere on the NSA during both typical conditions and periods of wildfire activity in North America, providing further evidence that terrestrial ecosystems and wildfires are an important source of aerosols in the Arctic. Additionally, results here indicate that the inland NSA may have more consistent impacts from boreal fires than the coastal sites.

3.3. IMPROVE Historical Record at Toolik

To contextualize aerosol observations during 2022 and 2023 analyzed in this study, we include complementary longer-term monitoring data from the IMPROVE network. The IMPROVE network (Malm et al., 1994) established an Air Quality monitoring site at Toolik Field Station in 2018. The 6-year record (late 2018–2023) is the only continuous record of air quality parameters such as OC and EC concentrations in the interior NSA (Figure 6). OC and EC concentrations reached maximums in the summer of 2022, coincident with wildfire smoke events. Mean OC was 800 ± 3,800 ng m^{-3} and mean EC was 180 ± 1,200 ng m^{-3} in 2022, about 3 times and 5 times

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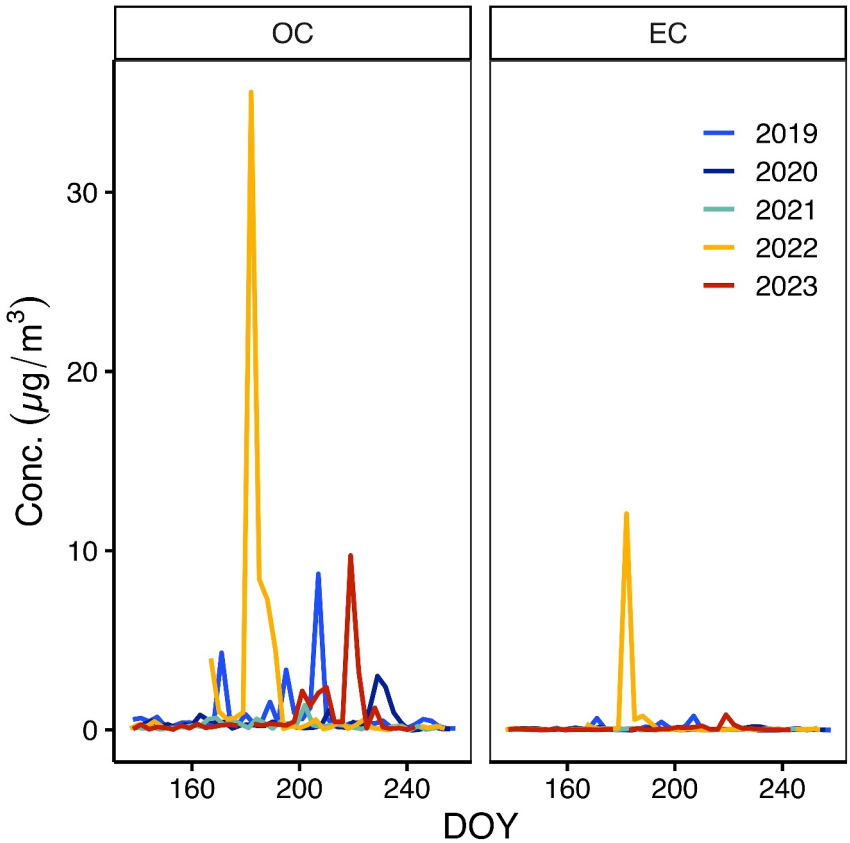

Figure 6. Concentrations of organic carbon (OC) and elemental carbon (EC) in fine particulate matter at Toolik from 2018 through July 2023, measured by the Interagency Monitoring of Protected Visual Environments (IMPROVE) program.

higher than the average of the previous 4 years combined, respectively. The next highest annual average OC and EC concentrations—420 and 50 ng m^{-3}, respectively—happened in 2023.

The IMPROVE record of OC and EC at Toolik (Figure 6) illustrates that the smoke captured in 2022 and 2023 resulted in remarkably high concentrations relative to the previous 4 years. The 2023 Canadian boreal forest fires —which were likely the major source of smoke to the NSA in 2023 (Figure S4 in Supporting Information S1)— were unprecedented, burning 10 times the average burned forest area in the last 40 years (Byrne et al., 2024). As changes to climate and weather patterns encourage more frequent and intense fires in the boreal region and even in the tundra, the increase in ozone, primary aerosol, and SOA precursors will change atmospheric chemistry in the Arctic (Heilman et al., 2014).

Monitoring through the IMPROVE network provides urgently needed context regarding current conditions on the NSA. As the climate changes rapidly, long-term observations will be critical for understanding resultant changes in atmospheric processes. As potential sources of carbonaceous aerosol to the region increase, the role of OC in summertime TSP will become increasingly important.

4. Conclusions

Our study provides insight into the character of summertime carbonaceous aerosols on the NSA. Using thermal-optical and ^{14}C analysis of TSP samples, we found significantly higher TC, OC, and EC concentrations in the interior NSA (Toolik site) compared to the Arctic coast (Utqiaġvik site). The dominance of contemporary over fossil sources indicates that natural sources—such as wildfire and biogenic precursors to SOA—largely controlled aerosol concentrations in Arctic Alaska during these summers. Moreover, our findings underscore the importance of boreal wildfire smoke events, specifically in North America, for carbonaceous aerosol concentrations. As the Arctic continues to warm and undergo ecological transformation, our work highlights the need

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for ongoing atmospheric monitoring efforts, especially in the low Arctic at sites with high terrestrial influence. The insight gained from such efforts will be critical in accurately predicting implications for atmospheric chemistry and radiative forcing.

Data Availability Statement

The data on total carbon, organic carbon, and elemental carbon concentrations, isotopic ratios, and estimates of source regions used in the study are published in the National Science Foundation's Office of Polar Programs Arctic Data Center repository (Welch et al., 2024).

Acknowledgments

This study was funded by the U.S. National Science Foundation Arctic Natural Sciences (OPP/ANS-2041251 to A.B.G. and C.I.C; ANS-2041250 to K.C.B., and ANS-2041240 to R.J.S) and U.S. Department of Energy Atmospheric Radiation Measurement field campaign AFC06964 for site support at Utqiaġvik. A.M.W. was supported by the Ridge to Reef NSF Research Traineeship, Award DGE-1735040. The authors thank the staff of the University of Alaska, Fairbanks and Toolik Field Station for logistic support, UC Irvine's KCCAMS facility for supporting elemental and isotope analyses, Ukpeaġvik Iñupiat Corporation (UIC) for field support at Utqiaġvik, DOE ARM for field support at Utqiaġvik, C. Moffett for preparation of samplers, J. Ivanoff (UIC) and P. Sheesley for sample collection at Utqiaġvik, and C. Leong (UC Irvine) for help with sample collection at Toolik. The authors acknowledge that Utqiaġvik and Toolik are the ancestral homeland of Inuit communities. Toolik Field Station and the surrounding areas are located on the ancestral hunting grounds of the Nunamiut, and occasional hunting grounds and routes of the Gwich'in, Koyukuk, and Iñupiaq peoples. Toolik Field Station—as well as the authors of this study—recognize and are grateful for the Indigenous people who inhabit and steward this land. The authors gratefully acknowledge the NOAA Air Resources Laboratory (ARL) for the provision of the HYSPLIT transport and dispersion model and/or READY website (https://www.ready.noaa.gov) used in this publication. IMPROVE is a collaborative association of state, tribal, and federal agencies, and international partners. The US Environmental Protection Agency is the primary funding source, with contracting and research support from the National Park Service. The Air Quality Group at the University of California, Davis is the central analytical laboratory, with ion analysis provided by Research Triangle Institute, and carbon analysis provided by Desert Research Institute.

References

Abbatt, J. P. D., Leaitch, W. R., Aliabadi, A. A., Bertram, A. K., Blanchet, J.-P., Boivin-Rioux, A., et al. (2019). Overview paper: New insights into aerosol and climate in the Arctic. *Atmospheric Chemistry and Physics*, *19*(4), 2527–2560. https://doi.org/10.5194/acp-19-2527-2019

Balshi, M. S., Mcguire, A. D., Duffy, P., Flannigan, M., Walsh, J., & Melillo, J. (2009). Assessing the response of area burned to changing climate in western boreal North America using a Multivariate Adaptive Regression Splines (MARS) approach. *Global Change Biology*, *15*(3), 578–600. https://doi.org/10.1111/j.1365-2486.2008.01679.x

Barrett, T. E., Robinson, E. M., Usenko, S., & Sheesley, R. J. (2015). Source contributions to wintertime elemental and organic carbon in the western Arctic based on radiocarbon and tracer apportionment. *Environmental Science & Technology*, *49*(19), 11631–11639. https://doi.org/10.1021/acs.est.5b03081

Barrett, T. E., & Sheesley, R. J. (2017). Year-round optical properties and source characterization of Arctic organic carbon aerosols on the North Slope Alaska. *Journal of Geophysical Research: Atmospheres*, *122*(17), 9319–9331. https://doi.org/10.1002/2016JD026194

Barrie, L. A. (1986). Arctic air pollution: An overview of current knowledge. *Atmospheric Environment*, *20*(4), 643–663. https://doi.org/10.1016/0004-6981(86)90180-0

Beverly, R. K., Beaumont, W., Tauz, D., Ormsby, K. M., von Reden, K. F., Santos, G. M., & Southon, J. R. (2010). The keck carbon cycle AMS laboratory, University of California, Irvine: Status Report. *Radiocarbon*, *52*(2), 301–309. https://doi.org/10.1017/S0033822200045343

Birch, M. E., & Cary, R. A. (1996). Elemental carbon-based method for monitoring occupational exposures to particulate diesel exhaust. *Aerosol Science and Technology*, *25*(3), 221–241. https://doi.org/10.1080/02786829608965393

Brutschin, E., & Schubert, S. R. (2016). Icy waters, hot tempers, and high stakes: Geopolitics and Geoeconomics of the Arctic. *Energy Research & Social Science*, *16*, 147–159. https://doi.org/10.1016/j.erss.2016.03.020

Byrne, B., Liu, J., Bowman, K. W., Pascolini-Campbell, M., Chatterjee, A., Pandey, S., et al. (2024). Carbon emissions from the 2023 Canadian wildfires. *Nature*, *633*(8031), 835–839. https://doi.org/10.1038/s41586-024-07878-z

Caiazzo, L., Calzolai, G., Becagli, S., Severi, M., Amore, A., Nardin, R., et al. (2021). Carbonaceous aerosol in polar areas: First results and improvements of the sampling strategies. *Atmosphere*, *12*(3), 320. https://doi.org/10.3390/atmos12030320

Castro, L. M., Pio, C. A., Harrison, R. M., & Smith, D. J. T. (1999). Carbonaceous aerosol in urban and rural European atmospheres: Estimation of secondary organic carbon concentrations. *Atmospheric Environment*, *33*(17), 2771–2781. https://doi.org/10.1016/S1352-2310(98)00331-8

Chen, Y., Romps, D. M., Seeley, J. T., Veraverbeke, S., Riley, W. J., Mekonnen, Z. A., & Randerson, J. T. (2021). Future increases in Arctic lightning and fire risk for permafrost carbon. *Nature Climate Change*, *11*(5), 1–7. https://doi.org/10.1038/s41558-021-01011-y

Christian, K. E., Brune, W. H., & Mao, J. (2017). Global sensitivity analysis of the GEOS-Chem chemical transport model: Ozone and hydrogen oxides during ARCTAS (2008). *Atmospheric Chemistry and Physics*, *17*(5), 3769–3784. https://doi.org/10.5194/acp-17-3769-2017

Chung, C. E., Ramanathan, V., & Decremer, D. (2012). Observationally constrained estimates of carbonaceous aerosol radiative forcing. *Proceedings of the National Academy of Sciences*, *109*(29), 11624–11629. https://doi.org/10.1073/pnas.1203707109

Corbett, J. J., Lack, D. A., Winebrake, J. J., Harder, S., Silberman, J. A., & Gold, M. (2010). Arctic shipping emissions inventories and future scenarios. *Atmospheric Chemistry and Physics*, *10*(19), 9689–9704. https://doi.org/10.5194/acp-10-9689-2010

Creamean, J. M., Maahn, M., de Boer, G., McComiskey, A., Sedlacek, A. J., & Feng, Y. (2018). The influence of local oil exploration and regional wildfires on summer 2015 aerosol over the North Slope of Alaska. *Atmospheric Chemistry and Physics*, *18*(2), 555–570. https://doi.org/10.5194/acp-18-555-2018

Dalsøren, S. B., Samset, B. H., Myhre, G., Corbett, J. J., Minjares, R., Lack, D., & Fuglestvedt, J. S. (2013). Environmental impacts of shipping in 2030 with a particular focus on the Arctic region. *Atmospheric Chemistry and Physics*, *13*(4), 1941–1955. https://doi.org/10.5194/acp-13-1941-2013

de Groot, W. J., Flannigan, M. D., & Cantin, A. S. (2013). Climate change impacts on future boreal fire regimes. *Forest Ecology and Management*, *294*, 35–44. https://doi.org/10.1016/j.foreco.2012.09.027

Deshmukh, D. K., Haque, M. M., Kim, Y., & Kawamura, K. (2019). Organic tracers of fine aerosol particles in central Alaska: Summertime composition and sources. *Atmospheric Chemistry and Physics*, *19*(22), 14009–14029. https://doi.org/10.5194/acp-19-14009-2019

Eleftheriadis, K., Vratolis, S., & Nyeki, S. (2009). Aerosol black carbon in the European Arctic: Measurements at Zeppelin station, Ny-Ålesund, Svalbard from 1998–2007. *Geophysical Research Letters*, *36*(2), L02809. https://doi.org/10.1029/2008GL035741

EPA, U. (2003). *Air quality Index: A guide to air quality and your health*. US Environmental Protection Agency.

Fisher, J. A., Jacob, D. J., Purdy, M. T., Kopacz, M., Le Sager, P., Carouge, C., et al. (2010). Source attribution and interannual variability of Arctic pollution in spring constrained by aircraft (ARCTAS, ARCPAC) and satellite (AIRS) observations of carbon monoxide. *Atmospheric Chemistry and Physics*, *10*(3), 977–996. https://doi.org/10.5194/acp-10-977-2010

Flannigan, M. D., Logan, K. A., Amiro, B. D., Skinner, W. R., & Stocks, B. J. (2005). Future area burned in Canada. *Climatic Change*, *72*(1), 1–16. https://doi.org/10.1007/s10584-005-5935-y

Garrett, T., Zhao, C., & Novelli, P. (2010). Assessing the relative contributions of transport efficiency and scavenging to seasonal variability in Arctic aerosol. *Tellus B: Chemical and Physical Meteorology*, *62*(3), 190–196. https://doi.org/10.1111/j.1600-0889.2010.00453.x

Gautier, D. L., Bird, K. J., Charpentier, R. R., Grantz, A., Houseknecht, D. W., Klett, T. R., et al. (2009). Assessment of undiscovered oil and gas in the Arctic. *Science*, *324*(5931), 1175–1179. https://doi.org/10.1126/science.1169467

Gillett, N. P., Weaver, A. J., Zwiers, F. W., & Flannigan, M. D. (2004). Detecting the effect of climate change on Canadian forest fires. *Geophysical Research Letters*, *31*(18), L18211. https://doi.org/10.1029/2004GL020876

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Gunsch, M. J., Kirpes, R. M., Kolesar, K. R., Barrett, T. E., China, S., Sheesley, R. J., et al. (2017). Contributions of transported Prudhoe Bay oil field emissions to the aerosol population in Utqiaġvik, Alaska. *Atmospheric Chemistry and Physics*, *17*(17), 10879–10892. https://doi.org/10.5194/acp-17-10879-2017

Hamilton, D. S., Lee, L. A., Pringle, K. J., Reddington, C. L., Spracklen, D. V., & Carslaw, K. S. (2014). Occurrence of pristine aerosol environments on a polluted planet. *Proceedings of the National Academy of Sciences*, *111*(52), 18466–18471. https://doi.org/10.1073/pnas.1415440111

Heilman, W. E., Liu, Y., Urbanski, S., Kovalev, V., & Mickler, R. (2014). Wildland fire emissions, carbon, and climate: Plume rise, atmospheric transport, and chemistry processes. *Forest Ecology and Management*, *317*, 70–79. https://doi.org/10.1016/j.foreco.2013.02.001

Hu, F. S., Higuera, P. E., Duffy, P., Chipman, M. L., Rocha, A. V., Young, A. M., et al. (2015). Arctic tundra fires: Natural variability and responses to climate change. *Frontiers in Ecology and the Environment*, *13*(7), 369–377. https://doi.org/10.1890/150063

Hwang, J., & Druffel, E. R. M. (2005). Blank correction for Δ14C measurements in organic compound classes of oceanic particulate matter. *Radiocarbon*, *47*(1), 75–87. https://doi.org/10.1017/S0033822200052218

Intergovernmental Panel on Climate Change (IPCC). (2022). Technical summary. In *The ocean and cryosphere in a changing climate: Special report of the intergovernmental panel on climate change* (pp. 39–70). Cambridge University Press. https://doi.org/10.1017/9781009157964.002

Kasischke, E. S., & Turetsky, M. R. (2006). Recent changes in the fire regime across the North American boreal region—Spatial and temporal patterns of burning across Canada and Alaska. *Geophysical Research Letters*, *33*(9), L09703. https://doi.org/10.1029/2006GL025677

Keeling, C. D. (1958). The concentration and isotopic abundances of atmospheric carbon dioxide in rural areas. *Geochimica et Cosmochimica Acta*, *13*(4), 322–334. https://doi.org/10.1016/0016-7037(58)90033-4

Kellomäki, S., Rouvinen, I., Peltola, H., Strandman, H., & Steinbrecher, R. (2001). Impact of global warming on the tree species composition of boreal forests in Finland and effects on emissions of isoprenoids. *Global Change Biology*, *7*(5), 531–544. https://doi.org/10.1046/j.1365-2486.2001.00414.x

Kelso, N. V., & Patterson, T. (2012). *5-Degree graticule grid, world, 1:110 million, 2012. Made with natural Earth*. Retrieved from http://www.naturalearthdata.com

Koch, D., & Hansen, J. (2005). Distant origins of Arctic black carbon: A Goddard Institute for Space Studies ModelE experiment. *Journal of Geophysical Research*, *110*(D4), D04204. https://doi.org/10.1029/2004JD005296

Kramshøj, M., Vedel-Petersen, I., Schollert, M., Rinnan, Å., Nymand, J., Ro-Poulsen, H., & Rinnan, R. (2016). Large increases in Arctic biogenic volatile emissions are a direct effect of warming. *Nature Geoscience*, *9*(5), 349–352. https://doi.org/10.1038/ngeo2692

Kristensen, D. K., Kristensen, E., Forchhammer, M. C., Michelsen, A., & Schmidt, N. M. (2011). Arctic herbivore diet can be inferred from stable carbon and nitrogen isotopes in C3 plants, faeces, and wool. *Canadian Journal of Zoology*, *89*(10), 892–899. https://doi.org/10.1139/z11-073

Law, K. S., & Stohl, A. (2007). Arctic air pollution: Origins and impacts. *Science*, *315*(5818), 1537–1540. https://doi.org/10.1126/science.1137695

Legendre, P. (2018). lmodel2: Model II regression. [Manual]. Retrieved From https://CRAN.R-project.org/package=lmodel2

Li, J., Carlson, B. E., Yung, Y. L., Lv, D., Hansen, J., Penner, J. E., et al. (2022). Scattering and absorbing aerosols in the climate system. *Nature Reviews Earth & Environment*, *3*(6), 363–379. https://doi.org/10.1038/s43017-022-00296-7

Li, X., Krueger, S. K., Strong, C., & Mace, G. G. (2020). Relationship between wintertime leads and low clouds in the Pan-Arctic. *Journal of Geophysical Research: Atmospheres*, *125*(18), e2020JD032595. https://doi.org/10.1029/2020JD032595

Lim, H.-J., & Turpin, B. J. (2002). Origins of primary and secondary organic aerosol in Atlanta: Results of time-resolved measurements during the Atlanta supersite experiment. *Environmental Science & Technology*, *36*(21), 4489–4496. https://doi.org/10.1021/es0206487

Lindwall, F., Schollert, M., Michelsen, A., Blok, D., & Rinnan, R. (2016a). Fourfold higher tundra volatile emissions due to arctic summer warming. *Journal of Geophysical Research: Biogeosciences*, *121*(3), 895–902. https://doi.org/10.1002/2015JG003295

Lindwall, F., Svendsen, S. S., Nielsen, C. S., Michelsen, A., & Rinnan, R. (2016b). Warming increases isoprene emissions from an arctic fen. *Science of The Total Environment*, *553*, 297–304. https://doi.org/10.1016/j.scitotenv.2016.02.111

Mackie, A. R., Palmer, P. I., Barlow, J. M., Finch, D. P., Novelli, P., & Jaeglé, L. (2016). Reduced Arctic air pollution due to decreasing European and North American emissions. *Journal of Geophysical Research: Atmospheres*, *121*(14), 8692–8700. https://doi.org/10.1002/2016JD024923

Malm, W. C., Sisler, J. F., Huffman, D., Eldred, R. A., & Cahill, T. A. (1994). Spatial and seasonal trends in particle concentration and optical extinction in the United States. *Journal of Geophysical Research*, *99*(D1), 1347–1370. https://doi.org/10.1029/93JD02916

Manousakas, M., Popovicheva, O., Evangeliou, N., Diapouli, E., Sitnikov, N., Shonija, N., & Eleftheriadis, K. (2020). Aerosol carbonaceous, elemental and ionic composition variability and origin at the Siberian High Arctic, Cape Baranova. *Tellus B: Chemical and Physical Meteorology*, *72*(1), 1–14. https://doi.org/10.1080/16000889.2020.1803708

McCarty, J. L., Aalto, J., Paunu, V.-V., Arnold, S. R., Eckhardt, S., Klimont, Z., et al. (2021). Reviews and syntheses: Arctic fire regimes and emissions in the 21st century. *Biogeosciences*, *18*(18), 5053–5083. https://doi.org/10.5194/bg-18-5053-2021

Moffett, C. E., Mehra, M., Barrett, T. E., Gunsch, M. J., Pratt, K. A., & Sheesley, R. J. (2022). Contemporary sources dominate carbonaceous aerosol on the North Slope of Alaska. *Science of The Total Environment*, *831*, 154641. https://doi.org/10.1016/j.scitotenv.2022.154641

Mori, T., Kondo, Y., Ohata, S., Zhao, Y., Sinha, P. R., Oshima, N., et al. (2020). Seasonal variation of wet deposition of black carbon in Arctic Alaska. *Journal of Geophysical Research: Atmospheres*, *125*(16), e2019JD032240. https://doi.org/10.1029/2019JD032240

Moschos, V., Dzepina, K., Bhattu, D., Lamkaddam, H., Casotto, R., Daellenbach, K. R., et al. (2022a). Equal abundance of summertime natural and wintertime anthropogenic Arctic organic aerosols. *Nature Geoscience*, *15*(3), 196–202. https://doi.org/10.1038/s41561-021-00891-1

Moschos, V., Schmale, J., Aas, W., Becagli, S., Calzolai, G., Eleftheriadis, K., et al. (2022b). Elucidating the present-day chemical composition, seasonality and source regions of climate-relevant aerosols across the Arctic land surface. *Environmental Research Letters*, *17*(3), 034032. https://doi.org/10.1088/1748-9326/ac444b

Mouteva, G. O., Czimczik, C. I., Fahrni, S. M., Wiggins, E. B., Rogers, B. M., Veraverbeke, S., et al. (2015). Black carbon aerosol dynamics and isotopic composition in Alaska linked with boreal fire emissions and depth of burn in organic soils. *Global Biogeochemical Cycles*, *29*(11), 1977–2000. https://doi.org/10.1002/2015GB005247

Mouteva, G. O., Fahrni, S. M., Santos, G. M., Randerson, J. T., Zhang, Y.-L., Szidat, S., & Czimczik, C. I. (2015). Accuracy and precision of 14C-based source apportionment of organic and elemental carbon in aerosols using the Swiss_4S protocol. *Atmospheric Measurement Techniques*, *8*(9), 3729–3743. https://doi.org/10.5194/amt-8-3729-2015

Muller, S. V., Racoviteanu, A. E., & Walker, D. A. (1999). Landsat MSS-derived land-cover map of northern Alaska: Extrapolation methods and a comparison with photo-interpreted and AVHRR-derived maps. *International Journal of Remote Sensing*, *20*(15–16), 2921–2946. https://doi.org/10.1080/014311699211543

NOAA. (2017). *Hazard mapping system (HMS) fire and smoke analysis* [Manual]. Office of Satellite and Product Operations, National Oceanic and Atmospheric Administration. Retrieved from http://www.ospo.noaa.gov/Products/land/fire.html

21698996, 2025, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JD042080, Wiley Online Library on [15/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Odwuor, A., Yañez, C. C., Chen, Y., Hopkins, F. M., Moreno, A., Xu, X., et al. (2023). Evidence for multi-decadal fuel buildup in a large California wildfire from smoke radiocarbon measurements. *Environmental Research Letters*, *18*(9), 094030. https://doi.org/10.1088/1748-9326/aced17

Pataki, D. E., Ehleringer, J. R., Flanagan, L. B., Yakir, D., Bowling, D. R., Still, C. J., et al. (2003). The application and interpretation of Keeling plots in terrestrial carbon cycle research. *Global Biogeochemical Cycles*, *17*(1), 1022. https://doi.org/10.1029/2001GB001850

Peters, G. P., Nilssen, T. B., Lindholt, L., Eide, M. S., Glomsrød, S., Eide, L. I., & Fuglestvedt, J. S. (2011). Future emissions from shipping and petroleum activities in the Arctic. *Atmospheric Chemistry and Physics*, *11*(11), 5305–5320. https://doi.org/10.5194/acp-11-5305-2011

Popovicheva, O., Diapouli, E., Makshtas, A., Shonija, N., Manousakas, M., Saraga, D., et al. (2019). East Siberian Arctic background and black carbon polluted aerosols at HMO Tiksi. *Science of The Total Environment*, *655*, 924–938. https://doi.org/10.1016/j.scitotenv.2018.11.165

Post, E., Alley, R. B., Christensen, T. R., Macias-Fauria, M., Forbes, B. C., Gooseff, M. N., et al. (2019). The polar regions in a 2°C warmer world. *Science Advances*, *5*(12), eaaw9883. https://doi.org/10.1126/sciadv.aaw9883

QGIS Development Team. (2023). *QGIS geographic information system [Manual]*. QGIS Association. Retrieved from https://www.qgis.org

Quinn, P. K., Bates, T. S., Baum, E., Doubleday, N., Fiore, A. M., Flanner, M., et al. (2008). Short-lived pollutants in the Arctic: Their climate impact and possible mitigation strategies. *Atmospheric Chemistry and Physics*, *8*(6), 1723–1735. https://doi.org/10.5194/acp-8-1723-2008

Quinn, P. K., Bates, T. S., Schulz, K., & Shaw, G. E. (2009). Decadal trends in aerosol chemical composition at Barrow, Alaska: 1976–2008. *Atmospheric Chemistry and Physics*, *9*(22), 8883–8888. https://doi.org/10.5194/acp-9-8883-2009

Quinn, P. K., Miller, T. L., Bates, T. S., Ogren, J. A., Andrews, E., & Shaw, G. E. (2002). A 3-year record of simultaneously measured aerosol chemical and optical properties at Barrow, Alaska. *Journal of Geophysical Research*, *107*(D11). https://doi.org/10.1029/2001JD001248

Quinn, P. K., Shaw, G., Andrews, E., Dutton, E. G., Ruoho-Airola, T., & Gong, S. L. (2007). Arctic haze: Current trends and knowledge gaps. *Tellus B: Chemical and Physical Meteorology*, *59*(1), 99–114. https://doi.org/10.1111/j.1600-0889.2006.00236.x

R Core Team. (2023). *R: A Language and environment for statistical computing*. R Foundation for Statistical Computing. Retrieved from https://www.R-project.org/

Ren, L., Yang, Y., Wang, H., Zhang, R., Wang, P., & Liao, H. (2020). Source attribution of Arctic black carbon and sulfate aerosols and associated Arctic surface warming during 1980–2018. *Atmospheric Chemistry and Physics*, *20*(14), 9067–9085. https://doi.org/10.5194/acp-20-9067-2020

Rinnan, R., Iversen, L. L., Tang, J., Vedel-Petersen, I., Schollert, M., & Schurgers, G. (2020). Separating direct and indirect effects of rising temperatures on biogenic volatile emissions in the Arctic. *Proceedings of the National Academy of Sciences*, *117*(51), 32476–32483. https://doi.org/10.1073/pnas.2008901117

Rodríguez, B. T., Huang, L., Santos, G. M., Zhang, W., Vetro, V., Xu, X., et al. (2020). Seasonal cycle of isotope-based source apportionment of elemental carbon in airborne particulate matter and snow at Alert, Canada. *Journal of Geophysical Research: Atmospheres*, *125*(23), e2020JD033125. https://doi.org/10.1029/2020JD033125

Rolph, G., Stein, A., & Stunder, B. (2017). Real-time environmental applications and display sYstem: Ready. *Environmental Modelling & Software*, *95*, 210–228. https://doi.org/10.1016/j.envsoft.2017.06.025

Schmale, J., Arnold, S. R., Law, K. S., Thorp, T., Anenberg, S., Simpson, W. R., et al. (2018). Local Arctic air pollution: A neglected but serious problem. *Earth's Future*, *6*(10), 1385–1412. https://doi.org/10.1029/2018EF000952

Schmale, J., Sharma, S., Decesari, S., Pernov, J., Massling, A., Hansson, H.-C., et al. (2022). Pan-Arctic seasonal cycles and long-term trends of aerosol properties from 10 observatories. *Atmospheric Chemistry and Physics*, *22*(5), 3067–3096. https://doi.org/10.5194/acp-22-3067-2022

Schmale, J., Zieger, P., & Ekman, A. M. L. (2021). Aerosols in current and future Arctic climate. *Nature Climate Change*, *11*(2), 95–105. https://doi.org/10.1038/s41558-020-00969-5

Schneider, E., Czech, H., Popovicheva, O., Chichaeva, M., Kobelev, V., Kasimov, N., et al. (2024). Mass spectrometric analysis of unprecedented high levels of carbonaceous aerosol particles long-range transported from wildfires in the Siberian Arctic. *Atmospheric Chemistry and Physics*, *24*(1), 553–576. https://doi.org/10.5194/acp-24-553-2024

Shapovalova, D. (2020). Arctic petroleum and the 2°C goal: A case for accountability for fossil-fuel supply. *Climate Law*, *10*(3–4), 282–307. https://doi.org/10.1163/18786561-10030003

Sharma, S., Andrews, E., Barrie, L. A., Ogren, J. A., & Lavoué, D. (2006). Variations and sources of the equivalent black carbon in the high Arctic revealed by long-term observations at Alert and Barrow: 1989–2003. *Journal of Geophysical Research*, *111*(D14), D14208. https://doi.org/10.1029/2005JD006581

Sharma, S., Ishizawa, M., Chan, D., Lavoué, D., Andrews, E., Eleftheriadis, K., & Maksyutov, S. (2013). 16-year simulation of Arctic black carbon: Transport, source contribution, and sensitivity analysis on deposition. *Journal of Geophysical Research: Atmospheres*, *118*(2), 943–964. https://doi.org/10.1029/2012JD017774

Sharma, S., Lavoué, D., Cachier, H., Barrie, L. A., & Gong, S. L. (2004). Long-term trends of the black carbon concentrations in the Canadian Arctic. *Journal of Geophysical Research*, *109*(D15), D15203. https://doi.org/10.1029/2003JD004331

Simin, T., Davie-Martin, C. L., Petersen, J., Høye, T. T., & Rinnan, R. (2022). Impacts of elevation on plant traits and volatile organic compound emissions in deciduous tundra shrubs. *Science of The Total Environment*, *837*, 155783. https://doi.org/10.1016/j.scitotenv.2022.155783

Sindelarova, K., Granier, C., Bouarar, I., Guenther, A., Tilmes, S., Stavrakou, T., et al. (2014). Global data set of biogenic VOC emissions calculated by the MEGAN model over the last 30 years. *Atmospheric Chemistry and Physics*, *14*(17), 9317–9341. https://doi.org/10.5194/acp-14-9317-2014

Stavroulas, I., Grivas, G., Michalopoulos, P., Liakakou, E., Bougiatioti, A., Kalkavouras, P., et al. (2020). Field evaluation of low-cost PM sensors (Purple Air PA-II) under variable urban air quality conditions, in Greece. *Atmosphere*, *11*(9), 926. https://doi.org/10.3390/atmos11090926

Stein, A. F., Draxler, R. R., Rolph, G. D., Stunder, B. J. B., Cohen, M. D., & Ngan, F. (2015). NOAA's HYSPLIT atmospheric transport and dispersion modeling system. *Bulletin of the American Meteorological Society*, *96*(12), 2059–2077. https://doi.org/10.1175/BAMS-D-14-00110.1

Stone, R. S., Sharma, S., Herber, A., Eleftheriadis, K., & Nelson, D. W. (2014). A characterization of Arctic aerosols on the basis of aerosol optical depth and black carbon measurements. *Elementa: Science of the Anthropocene*, *2*, 000027. https://doi.org/10.12952/journal.elementa.000027

Stuiver, M., & Polach, H. A. (1977). Discussion reporting of 14C data. *Radiocarbon*, *19*(3), 355–363. https://doi.org/10.1017/S0033822200003672

Sun, J., & Ariya, P. A. (2006). Atmospheric organic and bio-aerosols as cloud condensation nuclei (CCN): A review. *Atmospheric Environment*, *40*(5), 795–820. https://doi.org/10.1016/j.atmosenv.2005.05.052

Szidat, S., Jenk, T. M., Gäggeler, H. W., Synal, H.-A., Fisseha, R., Baltensperger, U., et al. (2004). Radiocarbon (14C)-deduced biogenic and anthropogenic contributions to organic carbon (OC) of urban aerosols from Zürich, Switzerland. *Atmospheric Environment*, *38*(24), 4035–4044. https://doi.org/10.1016/j.atmosenv.2004.03.066

21698996, 2025, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JD042080, Wiley Online Library on [15/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Tang, J., Zhou, P., Miller, P. A., Schurgers, G., Gustafson, A., Makkonen, R., et al. (2023). High-latitude vegetation changes will determine future plant volatile impacts on atmospheric organic aerosols. *Npj Climate and Atmospheric Science*, *6*(1), 147. https://doi.org/10.1038/s41612-023-00463-7

Valolahti, H., Kivimäenpää, M., Faubert, P., Michelsen, A., & Rinnan, R. (2015). Climate change-induced vegetation change as a driver of increased subarctic biogenic volatile organic compound emissions. *Global Change Biology*, *21*(9), 3478–3488. https://doi.org/10.1111/gcb.12953

Vettikkat, L., Miettinen, P., Buchholz, A., Rantala, P., Yu, H., Schallhart, S., et al. (2023). High emission rates and strong temperature response make boreal wetlands a large source of isoprene and terpenes. *Atmospheric Chemistry and Physics*, *23*(4), 2683–2698. https://doi.org/10.5194/acp-23-2683-2023

Walker, B. D., & Xu, X. (2019). An improved method for the sealed-tube zinc graphitization of microgram carbon samples and 14C AMS measurement. *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms*, *438*, 58–65. https://doi.org/10.1016/j.nimb.2018.08.004

Walker, X. J., Baltzer, J. L., Cumming, S. G., Day, N. J., Ebert, C., Goetz, S., et al. (2019). Increasing wildfires threaten historic carbon sink of boreal forest soils. *Nature*, *572*(7770), 520–523. https://doi.org/10.1038/s41586-019-1474-y

Wang, H., Welch, A., Nagalingam, S., Leong, C., Kittitananuvong, P., Barsanti, K. C., et al. (2024). Arctic heatwaves could significantly influence the isoprene emissions from shrubs. *Geophysical Research Letters*, *51*(2), e2023GL107599. https://doi.org/10.1029/2023gl107599

Wang, H., Welch, A. M., Nagalingam, S., Leong, C., Czimczik, C. I., Tang, J., et al. (2024). High temperature sensitivity of Arctic isoprene emissions explained by sedges. *Nature Communications*, *15*(1), 6144. https://doi.org/10.1038/s41467-024-49960-0

Wang, J. A., Baccini, A., Farina, M., Randerson, J. T., & Friedl, M. A. (2021). Disturbance suppresses the aboveground carbon sink in North American boreal forests. *Nature Climate Change*, *11*(5), 435–441. https://doi.org/10.1038/s41558-021-01027-4

Welch, A. M., Matthews, T., Sheesley, R. J., Wang, H., Barsanti, K. C., Nielsen, N., et al. (2024). Characterization of summertime carbonaceous aerosol on the North Slope of Alaska (Toolik and Utqiaġvik), 2022–2023 [Dataset]. *Arctic Data Center*. https://doi.org/10.18739/A20G3H09P

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., et al. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686

Willis, M. D., Leaitch, W. R., & Abbatt, J. P. D. (2018). Processes controlling the composition and abundance of Arctic Aerosol. *Reviews of Geophysics*, *56*(4), 621–671. https://doi.org/10.1029/2018RG000602

Winiger, P., Barrett, T. E., Sheesley, R. J., Huang, L., Sharma, S., Barrie, L. A., et al. (2019). Source apportionment of circum-Arctic atmospheric black carbon from isotopes and modeling. *Science Advances*, *5*(2), eaau8052. https://doi.org/10.1126/sciadv.aau8052

Xu, X., Trumbore, S. E., Zheng, S., Southon, J. R., McDuffee, K. E., Luttgen, M., & Liu, J. C. (2007). Modifying a sealed tube zinc reduction method for preparation of AMS graphite targets: Reducing background and attaining high precision. *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms*, *259*(1), 320–329. https://doi.org/10.1016/j.nimb.2007.01.175

Xu, Y., Zhuang, Q., Zhao, B., Billmire, M., Cook, C., Graham, J., et al. (2024). Impacts of wildfires on boreal forest ecosystem carbon dynamics from 1986 to 2020. *Environmental Research Letters*, *19*(6), 064023. https://doi.org/10.1088/1748-9326/ad489a

Yttri, K. E., Bäcklund, A., Conen, F., Eckhardt, S., Evangeliou, N., Fiebig, M., et al. (2023). Composition and sources of carbonaceous aerosol in the European Arctic at Zeppelin Observatory, Svalbard. *EGUsphere*, 1–49. https://doi.org/10.5194/egusphere-2023-615

Zhan, J., Gao, Y., Li, W., Chen, L., Lin, H., & Lin, Q. (2014). Effects of ship emissions on summertime aerosols at Ny–Alesund in the Arctic. *Atmospheric Pollution Research*, *5*(3), 500–510. https://doi.org/10.5094/APR.2014.059

Zhang, Y. L., Perron, N., Ciobanu, V. G., Zotter, P., Minguillón, M. C., Wacker, L., et al. (2012). On the isolation of OC and EC and the optimal strategy of radiocarbon-based source apportionment of carbonaceous aerosols. *Atmospheric Chemistry and Physics*, *12*(22), 10841–10856. https://doi.org/10.5194/acp-12-10841-2012

Zhong, Q., Schutgens, N., Veraverbeke, S., & van der Werf, G. R. (2024). Increasing aerosol emissions from boreal biomass burning exacerbate Arctic warming. *Nature Climate Change*, *14*(12), 1275–1281. https://doi.org/10.1038/s41558-024-02176-y

21698996, 2025, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2024JD042080, Wiley Online Library on [15/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License